

INTERNATIONAL STANDARD



**Wind energy generation systems –
Part 25-5: Communications for monitoring and control of wind power plants –
Compliance testing**



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INTERNATIONAL STANDARD



**Wind energy generation systems –
Part 25-5: Communications for monitoring and control of wind power plants –
Compliance testing**

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ELECTROTECHNICAL
COMMISSION

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

WIND ENERGY GENERATION SYSTEMS –**Part 25-5: Communications for monitoring
and control of wind power plants –
Compliance testing**

FOREWORD

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International Standard IEC 61400-25-5 has been prepared by IEC technical committee 88: Wind energy generation systems.

This second edition cancels and replaces the first edition published in 2006. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- Harmonization with structure and test cases in IEC 61850-10:2012.
- The use of SCL in the compliance testing process is out of the scope for this edition, but will be considered for Edition 3.
- Reduction of overlap between standards and simplification by increased referencing to the IEC 61850 standard series.

- All test cases applying SCL files are still not a part of the present document as the SCL specifications for wind power domain are still pending to be published.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
88/643/FDIS	88/650/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

Future standards in this series will carry the new general title as cited above. Titles of existing standards in this series will be updated at the time of the next edition.

A list of all parts of the IEC 61400 series, under the general title *Wind energy generation systems*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

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INTRODUCTION

The focus of IEC 61400-25 (all parts) is on the communications between wind power plant components such as wind turbines and actors such as SCADA Systems. Internal communication within wind power plant components is outside the scope of IEC 61400-25 (all parts).

IEC 61400-25 (all parts) is designed for a communication environment supported by a client-server model. Three areas are defined, that are modelled separately to ensure the scalability of implementations:

- a) wind power plant information models,
- b) information exchange model, and
- c) mapping of these two models to a standard communication profile.

The wind power plant information model and the information exchange model, viewed together, constitute an interface between client and server. In this conjunction, the wind power plant information model serves as an interpretation frame for accessible wind power plant data. The wind power plant information model is used by the server to offer the client a uniform, component-oriented view of the wind power plant data. The information exchange model reflects the whole active functionality of the server. IEC 61400-25 (all parts) enables connectivity between a heterogeneous combination of client and servers from different manufacturers and suppliers.

As depicted in Figure 1, IEC 61400-25 (all parts) defines a server with the following aspects:

- information provided by a wind power plant component, e. g., “wind turbine rotor speed” or “total power production of a certain time interval” is modelled and made available for access. The information modelled in the document is defined in IEC 61400-25-2,
- services to exchange values of the modelled information defined in IEC 61400-25-3,
- mapping to a communication profile, providing a protocol stack to carry the exchanged values from the modelled information (IEC 61400-25-4).

IEC 61400-25 (all parts) only defines how to model the information, information exchange and mapping to specific communication protocols. IEC 61400-25 (all parts) excludes a definition of how and where to implement the communication interface, the application program interface and implementation recommendations. However, the objective of IEC 61400-25 (all parts) is that the information associated with a single wind power plant component (such as the wind turbine) is accessible through a corresponding logical device.

The intended readers for the present document are device or system and/or system component manufacturers and test system developers/providers.

NOTE Abbreviations used in IEC 61400-25-5 are listed in Clauses 3 and 4 or can be found in other parts of IEC 61400-25 standard series that are relevant for compliance testing.

WIND ENERGY GENERATION SYSTEMS –

Part 25-5: Communications for monitoring and control of wind power plants – Compliance testing

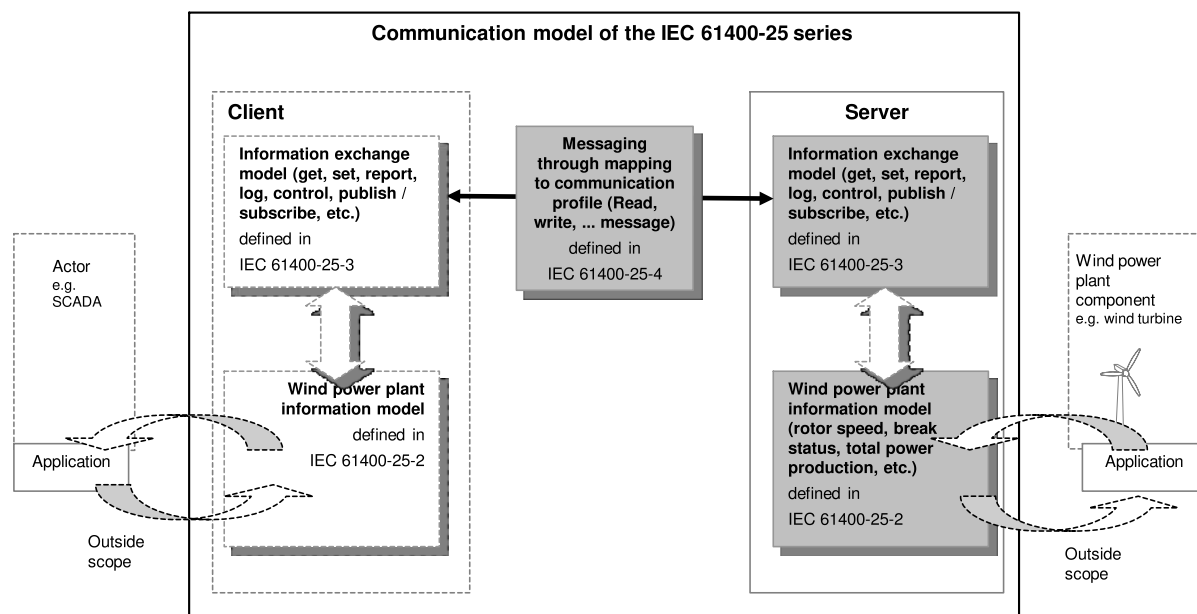
1 Scope

This part of IEC 61400-25 specifies standard techniques for testing of compliance of implementations, as well as specific measurement techniques to be applied when declaring performance parameters. The use of these techniques will enhance the ability of users to purchase systems that integrate easily, operate correctly, and support the applications as intended.

This part of IEC 61400-25 defines:

- the methods and abstract test cases for compliance testing of server and client devices used in wind power plants,
- the metrics to be measured in said devices according to the communication requirements specified in IEC 61400-25 (all parts).

NOTE The role of the test facilities for compliance testing and certifying the results are outside of the scope of IEC 61400-25-5.



IEC

Figure 1 – Conceptual communication model of the IEC 61400-25 standard series

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61400-25 (all parts), *Wind turbines – Part 25: Communications for monitoring and control of wind power plants*

IEC 61400-25-1:2006, *Wind turbines – Part 25-1: Communications for monitoring and control of wind power plants – Overall description of principles and models*

IEC 61400-25-2:2015, *Wind turbines – Part 25-2: Communications for monitoring and control of wind power plants – Information models*

IEC 61400-25-3:2015, *Wind turbines – Part 25-3: Communications for monitoring and control of wind power plants – Information exchange models*

IEC 61400-25-4:2016, *Wind energy generation systems – Part 25-4: Communications for monitoring and control of wind power plants – Mapping to communication profile*

IEC 61850-4:2011, *Communication networks and systems for power utility automation – Part 4: System and project management*

IEC 61850-6:2009, *Communication networks and systems for power utility automation – Part 6: Configuration description language for communication in electrical substations related to IEDs*

IEC 61850-7-1:2011, *Communication networks and systems for power utility automation – Part 7-1: Basic communication structure – Principles and models*

IEC 61850-7-2:2010, *Communication networks and systems for power utility automation – Part 7-2: Basic information and communication structure – Abstract communication service interface (ACSI)*

IEC 61850-7-3:2010, *Communication networks and systems for power utility automation – Part 7-3: Basic communication structure – Common data classes*

IEC 61850-7-4:2010, *Communication networks and systems for power utility automation – Part 7-4: Basic communication structure – Compatible logical node classes and data object classes*

IEC 61850-10:2012, *Communication networks and systems for power utility automation – Part 10: Conformance testing*

ISO/IEC 9646 (all parts), *Information technology – Open Systems Interconnection – Compliance testing methodology and framework*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 61400-25-1 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1

factory acceptance test

FAT

customer agreed functional tests of the specifically manufactured substation automation system or its parts using the parameter set for the planned application

Note 1 to entry: The FAT shall be carried out in the factory of the manufacturer or other agreed-upon location by the use of process simulating test equipment.

3.2

hold point

point, defined in the appropriate document beyond which an activity shall not proceed without the approval of the initiator of the compliance test

Note 1 to entry: The test facility shall provide a written notice to the initiator at an agreed time prior to the hold point.

Note 2 to entry: The initiator or his representative is obligated to verify the hold point and approve the proceeding of the activity.

3.3

interoperability

ability of two or more devices from the same vendor (or different vendors) to exchange information and use that information for correct co-operation

Note 1 to entry: Set of values defined corresponds with the quantities or values of another set.

3.4

model implementation compliance statement

MICS

statement that details the standard data object model elements supported by the system or device

3.5

negative test

test to verify the correct response of a device or a system when subjected to:

- IEC 61400-25 (all parts) conformant information and services which are not implemented in the device or system under test,
- non IEC 61400-25 (all parts) conformant information and services sent to the device or system under test

3.6

protocol implementation compliance statement

PICS

statement with the summary of the communication capabilities of the system or device to be tested

3.7

protocol implementation extra information for testing

PIXIT

statement with system or device specific information to be tested and which is outside the scope of IEC 61400-25 (all parts)

Note 1 to entry: The PIXIT is not subject to standardisation.

3.8

routine test

test performed by the manufacturer in order to ensure device operation and safety

3.9**site acceptance test****SAT**

verification of each data and control point and the correct functionality within the WPP and its operating environment at the whole installed plant by use of the final parameter set

Note 1 to entry: The SAT is the precondition for the WPP being put into operation.

3.10**system test**

verification of correct behaviour of the WPP components and of the overall WPP under various application conditions

Note 1 to entry: The system test marks the final stage of the development of a WPP system component.

3.11**system related test**

verification of correct behaviour of the IEDs and of the overall PUAS under specific application conditions

Note 1 to entry: The system related test is part of the final stage of the development of IEDs as belonging to a PUAS-product family.

3.12**test equipment**

all tools and instruments which simulate and verify the input/outputs of the operating environment of the WPP such as wind turbine, switchgear, transformers, network control centres or connected telecommunication units on the one side, and the communication links between the system components of the WPP on the other

3.13**test facility**

organisation able to provide appropriate test equipment and trained staff for compliance testing

Note 1 to entry: The management of compliance tests and the resulting information should follow a quality system.

3.14**technical issues compliance statement****TICS**

statement with device specific information regarding the implemented technical issues detected after publication of the standard

Note 1 to entry: The TICS is not subject to standardisation.

3.15**type test**

verification of correct behaviour of the IEDs (systems components) of the WPP by use of the system tested software under the test conditions corresponding with the technical data

Note 1 to entry: The type test marks the final stage of the hardware development and is the precondition for the start of the production. This test shall be carried out with system components which have been manufactured through the normal production cycle.

3.16**witness point**

point, defined in the appropriate document, at which an inspection will take place on an activity

Note 1 to entry: The activity may proceed without the approval of the initiator of the compliance test. The test facility provides a written notice to the initiator at an agreed time prior to the witness point. The initiator or his representative has the right, but is not obligated, to verify the witness point.

4 Abbreviated terms

ACSI	Abstract Communication Service Interface
BRCB	Buffered Report Control Block
CDC	Common Data Class
DUT	Device Under Test
FAT	Factory Acceptance Test
GI	General Interrogation
HMI	Human Machine Interface
ICD	IED capability subscription
IED	Intelligent Electronic Device
IID	Instantiated IED description
IP	Inter-networking protocol internet Protocol
LCB	Log Control Block
LD	Logical Device
LN	Logical Node
MICS	Model Implementation Compliance Statement
PICS	Protocol Implementation Compliance Statement
PIXIT	Protocol Implementation eXtra Information for Testing
PUAS	Power Utility Automation System
RCB	Report Control Block
RTU	Remote Terminal Unit
SAT	Site Acceptance Test
SCADA	Supervisory Control And Data Acquisition
SCSM	Specific Communication Service Mapping
SGCB	Setting Group Control Block
SoE	Sequence-of-Events
SUT	System Under Test
TICS	Technical Issues Compliance Statement
TPAA	Two Party Application Association
TUT	Tool Under Test
URCB	Unbuffered Report Control Block
UTC	Coordinated Universal Time
WPP	Wind Power Plant

5 Introduction to compliance testing

5.1 General

There are many steps involved from the development and production of a device to the proper running of a complete system designed according the specific needs of a customer. Suitable test steps are incorporated in this process.

Many internal tests during the development of a device or a system component result in a type test (unit level test) performed at least by the provider and – if required by applicable standards – by an independent test authority. In the context of this part of IEC 61400-25 (all parts), the term type test is restricted to the functional behaviour of the device excluding communication.

Continuing routine tests in the production chain are necessary to ensure a constant quality of delivered devices in accordance with the quality procedures of the producer.

A compliance test is the type test for communication and – since communication establishes a system – the basic integrated systems test of the incorporated system components. As a global communications standard, IEC 61400-25 (all parts) includes standardised compliance tests to ensure that all suppliers comply with applicable requirements.

Type tests and compliance tests do not completely guarantee that all functional and performance requirements are met. However, when properly performed, such tests significantly reduce the risk of costly problems occurring during system integration in the factory and on-site.

Compliance testing does not replace project specific system tests such as the FAT and SAT. The FAT and SAT are based on customer requirements for a dedicated WPP system and are done by the system integrator and normally witnessed by the customer. These tests increase the confidence level that all potential problems in the system have been identified and solved. These tests establish that the delivered WPP system is performing as specified.

5.2 Compliance test procedures

In general, compliance testing of the communication behaviour of a system component should address the functional requirements and performance requirements of typical applications supported by these devices in a WPP.

Compliance testing demonstrates the capability of the Device Under Test (DUT) to operate with other systems or system components in a specified way according to IEC 61400-25 (all parts).

Compliance testing requires consideration of the following issues:

- The problem of all testing is the completeness of the tests. The number of all possible situations can be very large. It may be possible to cover all normal operating cases, but this may not be true for all failure cases.
- It is impossible to test all system configurations using system components from different world-wide suppliers. Therefore, standardised test architecture with device simulators should be used. The use of such test architecture implies agreement about its configuration and the test procedures applied in order to achieve compatible results.
- A communication standard does not standardise the functions of the communicating equipment. Therefore, the failure modes of the functions are outside the scope of this part of IEC 61400-25 (all parts). But both the existence of distributed functions and the impact of function response in devices on the data flow create some interdependence.
- Depending on the definition range of IEC 61400-25 (all parts), some properties of the device may be proven by information and documents provided with the DUT for the compliance testing instead of the compliance test itself.

The compliance test establishes that the communication of the DUT works according to IEC 61400-25 (all parts). IEC 61400-25 (all parts) is focused on interoperability using data objects, functions and device models including services above or at the application level (ACSI).

Since IEC 61400-25 (all parts) defines no new communication stacks, the compliance to all seven ISO/OSI layers may be proven by documentation that communication stack software compliant with the corresponding specifications is implemented and may have been pre-tested and optionally certified. In the standard compliance test, only the application according to ACSI can be tested.

5.3 Quality assurance and testing

5.3.1 General

In order to assure the quality during compliance testing, a quality assurance system has to be in place. This shall be clearly demonstrated by the test facility and apply as well to the quality assurance system of any sub-supplier.

In general, quality surveillance is used to monitor and verify the status of components during all phases of the compliance tests. For this purpose, inspections are carried out, based on hold and witness points that are indicated by the initiator or its representative in the test and the inspection plan that is supplied by the test facility. These inspections are process-related and will provide information and confidence on the quality of the tests. Quality surveillance will reduce the risks of failure during the FAT and SAT.

5.3.2 Quality plan

5.3.2.1 Compliance test quality plan

The test facility will supply, for evaluation, a quality plan for the compliance test.

The plan shall describe all measures for the scope of work and/or deliveries in the areas of organisation, time, information and quality. There is only one plan for the test facility and its sub-suppliers.

The compliance test quality plan is proposed to contain the following:

- A complete and detailed description of the work methods. This will help ensure that all verifiable activities will fulfil all applicable requirements and conditions as stated in the scope of work during the time allowed.
- A detailed description of all tasks to be performed, including references to the schedule, an overview of the involved staff, materials and work methods as well as relevant methods and procedures.
- A detailed description of the organisation, including the assignments, tasks and responsibilities of mentioned staff during the different stages of the test programs. The description shall include all tests, inspections, research and audits during the various stages of the tests and the dates on which they will take place. These descriptions will be part of the test and inspection plan.
- A method for handling deviations, changes and modifications during all stages of the test.
- A sign off procedure and a description of the documentation to be supplied.

5.3.2.2 Test and inspection plan

The compliance test quality plan shall contain a test and inspection plan. In this plan, the test facility specifies, for all phases of the tests:

- what will be inspected, tested and registered;
- the purpose of the inspections and tests;
- the procedures and standards to which inspections, tests and registrations will be performed;
- the expected results of the inspections and tests;
- identification of persons to perform the inspections, tests and registrations.

The test facility is responsible for the correct and timely performance of all activities mentioned in the test and inspection plan.

The test facility shall provide a proposal for so called hold, witness and review points in the test and inspection plan.

There are several methods to perform a hold or witness point. The initiator of the compliance test or a representative can be present during the execution of a test or inspection. It is also possible to review the associated quality documents, e.g. checklists, verification and validation documents. The hold or witness point can take place at the test facilities site during the execution of a test, or an inspection can be made at the initiator's site in which case the test facilitator shall provide all relevant documents to the initiator.

All hold and witness points will be announced by the test facility at least a predefined time before they take place. A period of at least one week is recommended, depending on the time needed for making travel arrangements and the availability of the needed resources.

5.3.2.3 Audits requested by initiator

The initiator of a compliance test has the right to conduct audits on the quality system of the test facility and its sub-suppliers. The test facility shall co-operate and provide access to all locations applicable for the compliance test. The initiator's right to check that the quality of the compliance test does not dismiss the test facility from its responsibilities.

Inspections and tests by the initiator of a compliance test shall be possible at mutually agreeable times at the locations, offices and factories of the test facility and all applicable third parties and sub-suppliers.

5.4 Testing

5.4.1 General

Compliance testing shall be customised for each device under test based on the capabilities identified in the PICS, PIXIT and MICS provided by the vendor. When submitting devices for testing, the following shall be provided:

- device ready for testing;
- protocol implementation compliance statement (PICS);
- protocol implementation extra information for testing (PIXIT) statement;
- model implementation compliance statement (MICS); technical issues compliance statement (TICS);
- instruction manuals detailing the installation and operation of the device.

The requirements for compliance testing fall into two categories:

- static compliance requirements (defines the requirements the implementation shall fulfil);
- dynamic compliance requirements (defines the requirements that arise from the protocol used for a certain implementation).

The static and dynamic compliance requirements shall be defined in a protocol implementation compliance statement or PICS. The PICS serves three purposes:

- a) selection of the appropriate set of tests;
- b) ensure that the tests appropriate to a claim of compliance are performed;
- c) provide the basis for the review of the static compliance.

A standard PICS shall be provided.

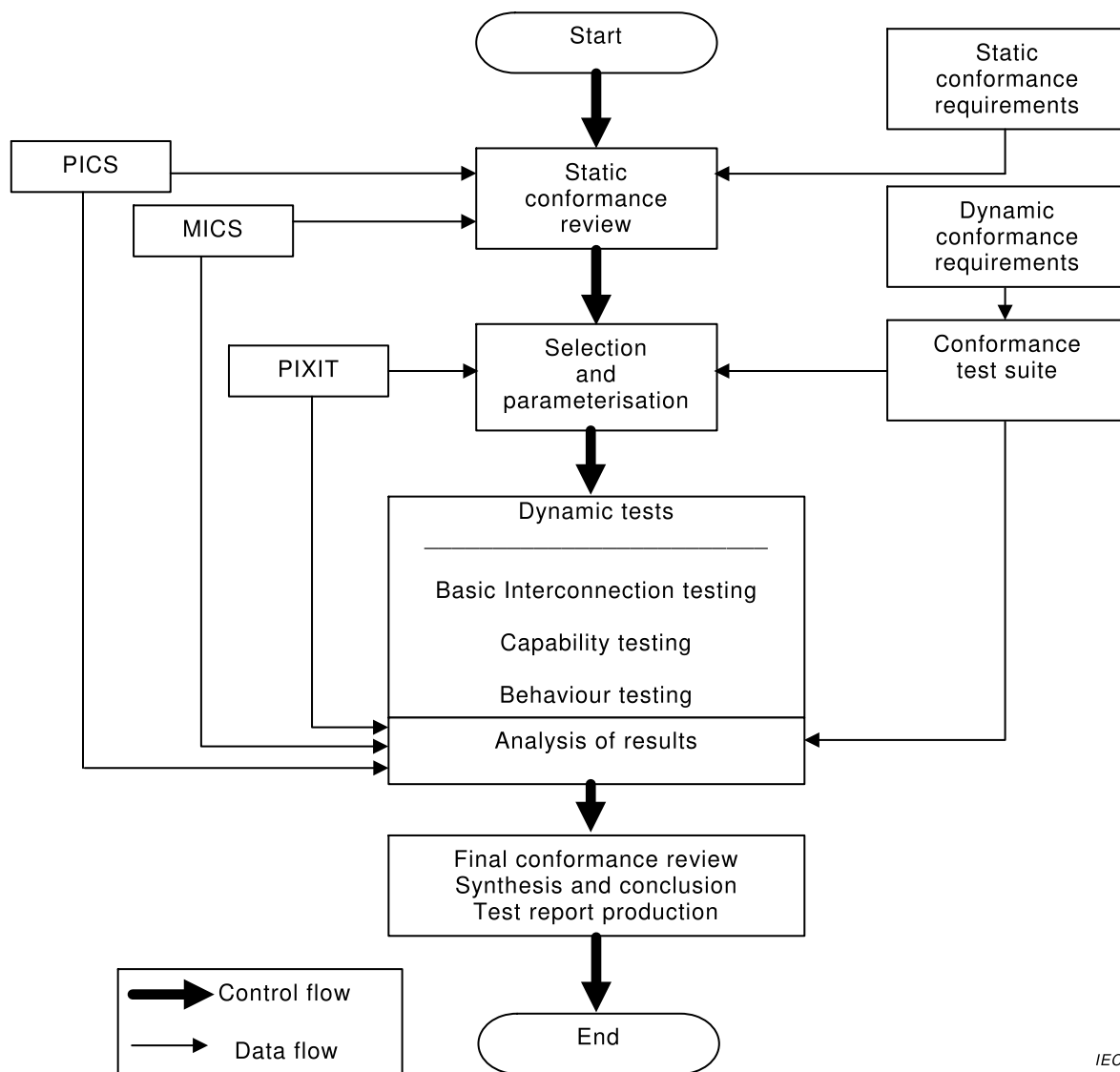
Concrete PICS shall be as defined for the SCSMs.

A model implementation compliance statement or MICS shall be provided detailing the standard data object model elements supported by the system or device. The MICS is implemented in the ICD of IID file according to IEC 61850-6.

A technical issues compliance statement (TICS) shall be provided detailing the implemented technical issues after publication of the standard.

In addition to the PICS, a protocol implementation extra information for testing (PIXIT) document shall be provided.

The process of assessing the compliance is shown in Figure 2.



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Figure 2 – Conceptual compliance assessment process

5.4.2 Device testing

A single device shall be compliance tested against a test device.

The device-specific compliance tests contain the positive and negative testing of the following as appropriate:

- inspection of the documentation and version control of the device,
- test of device configuration file against the device related object model (IEC 61400-25-2),
- test of communication stack implementation against applicable SCSM (IEC 61400-25-4),
- test of implemented ACSI services against ACSI definition (IEC 61400-25-3),

- test of device specific extensions according to rules given by IEC 61400-25 (all parts) in general.

5.5 Documentation of compliance test report

A compliance test report shall include the following information:

- A reference list of all documents that describe or specify any qualifying tests that have been performed. These documents may include the vendor's standard operating and testing procedures, and local, national and international standards. International standards shall be cited by document number, date, clauses and subclauses. References to other documents shall include a complete source address and document identification. A complete and contextually accurate summary or extract of the document may be included for convenience.
- A list of any specialised test equipment or computer programs used for performing the compliance tests.
- Name and address of the vendor.
- Name and address of the initiator of the compliance test (if different from vendor name).
- Name of the tested device.
- All of the variants (hardware, firmware, etc.) of tested device.
- Name and address of the test facility.
- Date of issue of test report.
- Name and signature of the tester.
- Unique reference number.
- A list of test items performed to verify compliance.
- Comments and problems found.
- For each test item, the following subjects shall be documented:
 - description of the test item with the objective of the test, the test procedure and the expected result;
 - reference to IEC 61400-25 series (all parts), clause and subclause;
 - unique identifier per test item;
 - test result: passed, failed, inconclusive, not applicable;
 - comparison of the test result to the expected result.

Changes or alterations to the device made at any point in the test, particularly those made to correct a test deficiency, shall be completely described. The consequences and requirements of re-testing of a device – if required – shall be specified in corresponding test plans and test reports.

Compliance test documentation shall be supplied to the initiator.

6 Device related compliance testing

6.1 Test methodology

Communication testing needs at least two devices to communicate with each other. Comprehensive interoperability testing of all possible products is not feasible. Therefore, the test concept shall include test devices, test configurations, and test scenarios.

The dynamic behaviour should be tested properly by using well-defined test cases.

Special attention shall be given to communication equipment such as star-couplers, switches, etc., which shall support all requested features of IEC 61400-25 (all parts), but not introduce

additional contingencies and limitations. The impact of the communication method (client-server, FTP/IP, etc.) used by the device under test shall be considered properly in the test procedures. Verification of functional applications is not part of a compliance test even if advanced tools may offer such analysis.

6.2 Compliance test procedures

6.2.1 General

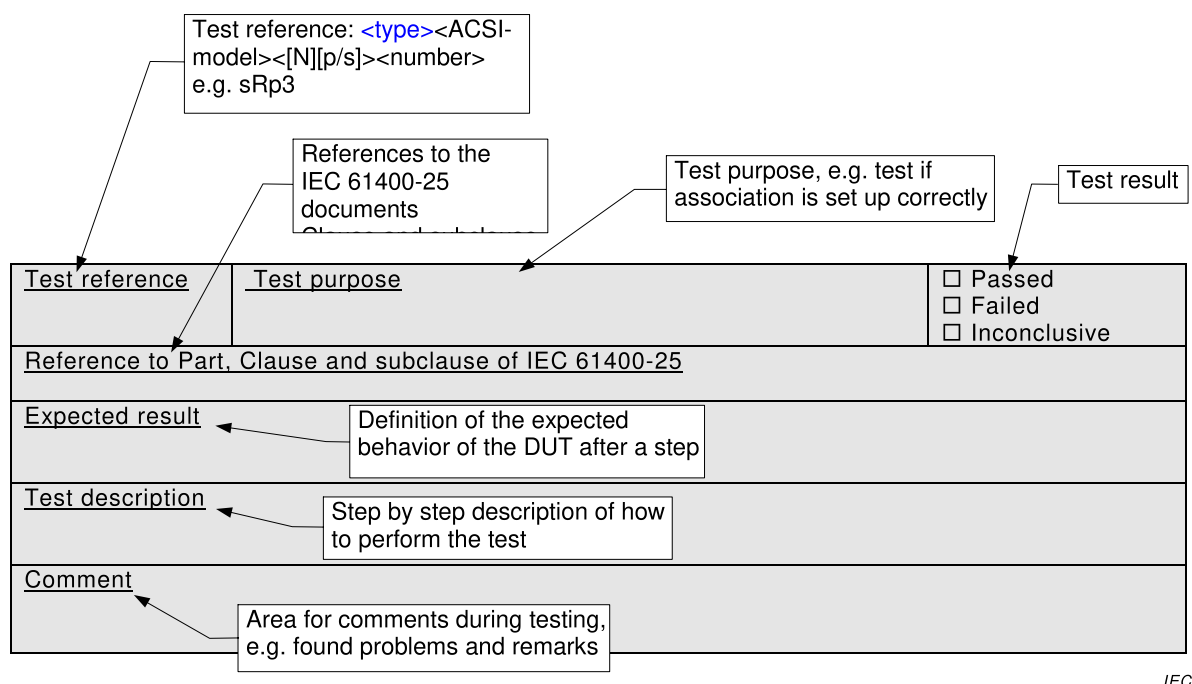
Subclause 6.2 describes the test procedure requirements, test structure, the abstract test cases (what is to be tested). The format and two examples of test procedures (how it is to be tested) are given in Annex A.

6.2.2 Test procedure requirements

The test procedure requirements are:

- The abstract test cases describe what shall be tested; the test procedures describe how a test engineer or a test system shall perform the test.
- Test cases include a reference to the applicable paragraph(s) in the referenced document(s).
- The test results shall be reproducible in the same test lab and in other test labs.
- Support automated testing with minimal human intervention, as far as reasonably possible.
- The tests shall focus on situations that cannot easily be tested during, for example, a factory or site acceptance test, and prevent inter-operability risks, for example:
 - check behaviour of the device on delayed, lost, double and out of order packets,
 - configuration, implementation, operation risks,
 - mismatching names, parameters, settings, or data types,
 - exceeding certain limits, ranges or timeouts,
 - force situations to test negative responses,
 - check all (control) state machine paths, and
 - force simultaneous control operations from multiple clients.
- The ACSI tests focus on the application layer (mapping).
- The device under test (DUT) is considered as a black box. The I/O and the communication interface are used for testing.
- The test includes testing the versions, data model and configuration file, and the use of applicable ISO/IEC 9646 series terminology.

The test procedures shall be formatted as outlined in Figure 3. With this format, the test procedures document can also be used as test report. A few test procedure examples are given in Annex A.

**Figure 3 – Test procedure format**

6.2.3 Test structure

The server test cases are structured as follows:

- Documentation and version control (IEC 61400-25-5).
- Data model (IEC 61400-25-2).
- Mapping of ACSI models and services (IEC 61400-25-3).
- All test cases are inherited from IEC 61850-10:2012 where applicable for IEC 61400-25 (all parts).

6.2.4 Test cases to test a server device

6.2.4.1 General

This part of IEC 61400-25 specifies the test system architecture and abstract test cases for a server device. The abstract test cases shall be used for the definition of test procedures to run in tests.

The SCSM specific test procedures shall be provided by test facilities agreed upon by the market participants

6.2.4.2 Test system architecture to test a server device

In order to be able to perform a device test, a minimum test set-up is necessary. The test architecture contains (see Figure 4):

- the DUT;
- a client simulator to initiate and generate TPAA messages and record and process resulting information;
- a master for time synchronisation (the time sync master);
- engineering tool software to configure the DUT;
- a protocol/network analyser to monitor and store all the network traffic for each test case;

- a signal generator to force binary and analogue events, controlled by the test master or test engineer.

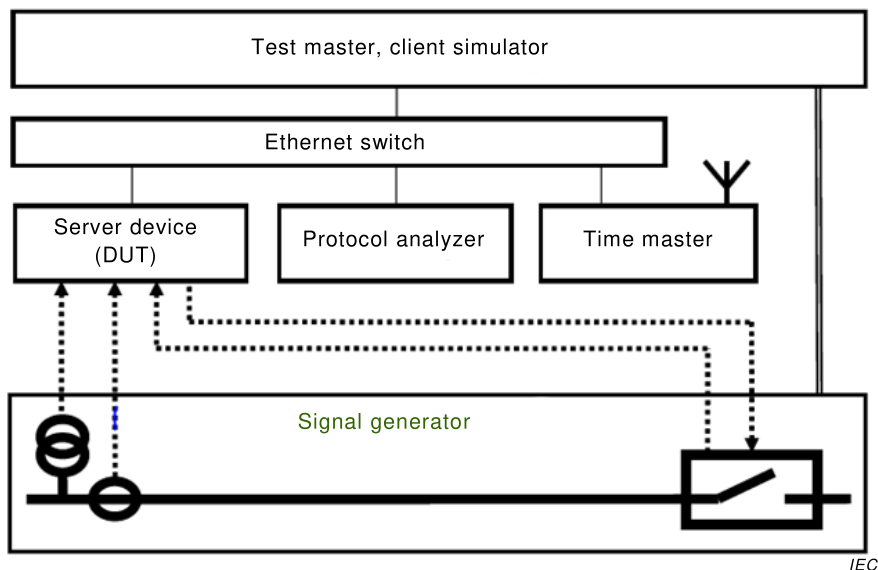


Figure 4 – Test system architecture to test a server device

The test system shall include documentation regarding test system hardware and test system software.

6.2.4.3 Documentation and version control test procedure overview

The test cases listed in Table 1 shall apply.

Table 1 – Server documentation test cases

Test case	Test case description
sDoc1	Check if the major/minor software version in the PICS documentation and the DUT do match (IEC 61850-4). PICS shall contain the ACSI compliance statement according to IEC 61400-25-3
sDoc2	Check if the major/minor software version manufacturer PIXIT documentation and software version of the DUT do match (IEC 61850-4). PIXIT shall indicate the required information as requested in the test cases
sDoc3	Check if the major/minor software version manufacturer MICS documentation and software version of the DUT do match (IEC 61850-4). MICS shall indicate the semantics of all non-standard Logical Nodes, Data Objects, Data Attributes and enumeration
sDoc4	Check if the major/minor software version manufacturer TICS documentation and software version of the DUT do match (IEC 61850-4). TICS shall indicate the implemented technical issues

6.2.4.4 Data model test cases

The test cases listed in Table 2 shall apply.

Table 2 – Server data model test cases

Test case	Test case description
sMdl1	Verify presence of mandatory objects for each LN. Passed when all objects/attributes are present
sMdl2	Verify presence of conditional presence true objects for each LN. Passed when all objects/attributes are present
sMdl3	Verify non-presence of conditional presence false objects. Passed when these objects/attributes are not present
sMdl4	Verify data model mapping according to applicable SCSM concerning name length and object expansion. Passed when mapping is according to applicable SCSM
sMdl5	Verify data model mapping according to applicable SCSM concerning organisation of functional components. Passed when mapping is according to applicable SCSM
sMdl6	Verify data model mapping according to applicable SCSM concerning naming of control blocks and logs. Passed when mapping is according to applicable SCSM
sMdl7	Verify data type of all objects for each LN. Passed when data type of all objects/attributes do match with IEC 61400-25-2, IEC 61400-25-3 and the applicable SCSM
sMdl8	Verify data attribute values from the device are in specified range (this is a continuous effort during the whole compliance test). Passed when all values are in range
sMdl9	Check if manufacturer specific data model extensions are implemented according to the extension rules in IEC 61400-25-2 (only when extensions are implemented). Passed when all extensions are implemented according to the specified rules
sMdl10	Check if the order of the data attributes is within the Data Object types match with IEC 61400-25-2. Passed when all attributes are in matching order
sMdl11	Check the maximum name length of Logical Device, Logical Node, data sets and control blocks according to IEC 61850-7-2:2010, 22.2 and the applicable SCSM.
sMdl12	Check that the rules for multiple data object instantiation are kept (IEC 61850-7-1, IEC 61850-7-4).
sMdl13	Check that the logical device name space refers to IEC 61400-25-2:2015.
sMdl14	Check the correct use of name spaces for non-WPP applications.

6.2.4.5 Mapping of ACSI models and services test cases

Test items shall be grouped together in tables. The tables shall reflect the applicable service models specified in IEC 61850-7-2:2010, Figure 3:

- Application association (sAss);
- Server, Logical device, Logical node, Data, and Data Attribute model (sSrv);
- Data set model (sDs);
- Unbuffered report control model (sRp);
- Buffered report control model (sBr);
- Log control model (sLog);
- Control model (sCtl);
- Time synchronisation model (sTm).

Test cases are defined for each ACSI model and services in the following categories:

- positive = verification of normal conditions, typically resulting in response+;
- negative = verification of abnormal conditions, typically resulting in response–.

A test case is mandatory when the applicable ACSI model and ACSI service is supported by the DUT. This is specified in the PICS according to IEC 61850-7-2:2010, Annex A. The test result interpretation (passed/failed) depends on the declared IED capabilities e.g. in the ICD file as well as on the test result.

6.2.4.6 Application association

6.2.4.6.1 Positive test cases

The test cases listed in Table 3 shall apply.

Table 3 – Association positive test cases

Test case	Test case description
sAss1	Associate and release a TPAA association (IEC 61850-7-2:2010, 8.3).
sAss2	Associate and client-abort TPAA association (IEC 61850-7-2:2010, 8.3).
sAss3	Associate with maximum number of clients simultaneously (PIXIT).

6.2.4.6.2 Negative test cases

The test cases listed in Table 4 shall apply.

Table 4 – Association negative test cases

Test case	Test case description
sAssN1	Check that with incorrect authentication parameters and authentication turned on at server, the association fails, and with authentication turned off, the server associates (IEC 61850-7-2:2010, 8.3).
sAssN2	Check that with incorrect association parameters at server or client the association fails (IEC 61850-7-2:2010, 8.3, PIXIT).
sAssN3	Set up maximum+1 associations, verify the last associate is refused.
sAssN4	Disconnect the communication interface, the DUT shall detect link lost within a specified period.
sAssN5	Interrupt and restore the power supply, the DUT shall accept an association request when ready.
sAssN6	Verify the re-use of dropped association resources

6.2.4.7 Server, Logical Device, Logical Node, and Data model

6.2.4.7.1 Positive test cases

The test cases listed in Table 5 shall apply.

Table 5 – Server positive test cases

Test case	Test case description
sSrv1	Request GetServerDirectory(LOGICAL-DEVICE) and check response (IEC 61850-7-2:2010, 7.2.2)
sSrv2	For each GetServerDirectory(LOGICAL-DEVICE) response issue a GetLogicalDeviceDirectory request and check response (IEC 61850-7-2:2010, 9.2.1)
sSrv3	For each GetLogicalDeviceDirectory response issue a GetLogicalNodeDirectory(DATA) request and check response (IEC 61850-7-2:2010, 10.2.2)
sSrv4	For each GetLogicalNodeDirectory(DATA) response issue a – GetDataDirectory request and check response (IEC 61850-7-2:2010, 11.4.4) – GetDataDefinition request and check response (IEC 61850-7-2:2010, 11.4.5) – GetDataValues request and check response (IEC 61850-7-2:2010, 11.4.2)
sSrv5	Issue one GetDataValues request with the maximum number of data values and check response
sSrv6	For each write enabled DATA object issue a SetDataValues request and check response (IEC 61850-7-2:2010, 11.4.3)
sSrv7	Issue one SetDataValues request with the maximum number of data values and check response
sSrv8	Request GetAllDataValues for each functional constraint and check response (IEC 61850-7-2:2010, 10.2.3)
sSrv9	Evaluate the semantic of selected (volt/amp) analogue measurements: – Verify analogue value (plausibility check, not accuracy) – Verify quality bits, force situations to set specific quality bits – Verify (UTC) timestamp value and quality (plausibility check, not accuracy) – Verify scaling, range and units, change a setting and verify resulting value – Verify dead band, change dead band and verify result – Verify limit indications
sSrv10	Evaluate the semantic of selected status points: – Verify status value – Verify quality bits, force situations to set specific quality bits – Verify (UTC) timestamp value and quality (plausibility check, not accuracy)
sSrv11	Verify that when blkEna is set to true by an operator the quality bit oldData and operatorBlocked is set by the server and the process data is not updated anymore; by an (internal) automatic function the quality oldData is set and the process data is not updated anymore (IEC 61850-7-3:2010, 6.2.6)
sSrv12	Verify Mod/Beh values: off, test, blocked – When Mod/Beh is off process data is not updated, Mod and Beh are updated, quality is set to invalid – When Mod/Beh is test or test-blocked the process data quality test is set – When Mod/Beh is on-blocked the process data quality operatorBlocked is set – (IEC 61850-7-4:2010, Annex A)
sSrv13	Verify logical device hierarchy; – the LLN0.GrRef should reference a valid logical device – the reference shall not result in a hierarchy loop – Beh value at higher level influences the lower levels correctly (i.e. like LD Beh influences LN behaviour dependent on LN Mod)

6.2.4.7.2 Negative test cases

The test cases listed in Table 6 shall apply.

Table 6 – Server negative test cases

Test case	Test case description
sSrvN1	Request the following data services with wrong parameters (unknown object, name case mismatch, wrong logical device or wrong logical node) and verify response-service error – ServerDirectory(LOGICAL-DEVICE) (IEC 61850-7-2:2010, 7.2.2), – GetLogicalDeviceDirectory (IEC 61850-7-2:2010, 9.2.1), – GetLogicalNodeDirectory(DATA) (IEC 61850-7-2:2010, 10.2.2), – GetAllDataValues (IEC 61850-7-2:2010, 10.2.3), – GetDataValues (IEC 61850-7-2:2010, 11.4.2), – SetDataValues (IEC 61850-7-2:2010, 11.4.3), – GetDataDirectory (IEC 61850-7-2:2010, 11.4.4), – GetDataDefinition (IEC 61850-7-2:2010, 11.4.5).
sSrvN2	Request SetDataValues of ENUMERATED data with out-of-range value and verify response- service error (IEC 61850-7-2:2010, 11.4.3).
sSrvN3	Request SetDataValues with mismatching data type (e.g. int-float) and verify response- service error (IEC 61850-7-2:2010, 11.4.3).
sSrvN4	Request SetDataValues for read-only data values and verify response- service error (IEC 61850-7-2:2010, 11.4.3).

6.2.4.8 Data set model

6.2.4.8.1 Positive test cases

The test cases listed in Table 7 shall apply.

Table 7 – Data set positive test cases

Test case	Test case description
sDs1	Request GetLogicalNodeDirectory(LOGICAL-DEVICE) and check response (IEC 61850-7-2:2010, 10.2.2). For each response, issue a: – GetDataSetValues request and check response (IEC 61850-7-2:2010, 13.3.2), – GetDataSetDirectory request and check response (IEC 61850-7-2:2010, 13.3.6).
sDs2	Request a persistent CreateDataSet with one member and with maximum possible members and check response (IEC 61850-7-2:2010, 13.3.4) and verify that the non-persistent data set is visible for another client.
sDs3	Request a non-persistent CreateDataSet with one member and with maximum possible members and check response (IEC 61850-7-2:2010, 13.3.4) and verify that the persistent data set is not visible for another client.
sDs4	Create and delete a persistent dataset, create the dataset again with the same name with one extra data value/re-ordered member and check the members.
sDs5	Create and delete a non-persistent dataset, create the dataset again with the same name with one extra data value/re-ordered member and check the members.
sDs6	Create a non-persistent dataset, release/abort the association, associate again and check that the dataset has been deleted (IEC 61850-7-2:2010, 13.1).
sDs7	Create a non-persistent dataset, release/abort the association, associate again and check that the dataset is still present (IEC 61850-7-2:2010, 13.1).
sDs8	Create and delete a persistent dataset and verify that every data set can be created normally: repeat the process of creating and deleting once.
sDs9	Create and delete a non-persistent dataset and verify that every data set can be created normally: repeat the process of creating and deleting once.
sDs10	Verify SetDataSetValues/GetDataSetValues with GetDataValues and SetDataValues.
sDs11	Verify that the maximum number of persistent data sets with the maximum number of members can be created for each of the specified maximum number of clients as specified
sDs12	Verify that the maximum number of non-persistent data sets with the maximum number of members can be created for each of the specified maximum number of clients as specified
sDs13	Verify that a non-persistent data set can be created with the maximum name length for data set and a data set member (IEC 61850-7-2:2010, 22.2)
sDs14	Verify that a persistent data set can be created with the maximum name length for data set and a data set member (IEC 61850-7-2:2010, 22.2)

6.2.4.8.2 Negative test cases

The test cases listed in Table 8 shall apply.

Table 8 – Data set negative test cases

Test case	Test case description
sDsN1	Request the following data set services with wrong parameters (unknown object, name case mismatch, wrong logical device or wrong logical node) and verify response-service error – GetDataSetValues (IEC 61850-7-2:2010, 13.3.2) – SetDataSetValues (IEC 61850-7-2:2010, 13.3.3) – CreateDataSet (IEC 61850-7-2:2010, 13.3.4) – DeleteDataSet (IEC 61850-7-2:2010, 13.3.5) – GetDataSetDirectory (IEC 61850-7-2:2010, 13.3.6)
sDsN2	Create a persistent dataset with the same name twice, and verify response-service error
sDsN3	Create a non-persistent dataset with the same name twice, and verify response-service error
sDsN4	Continue to create persistent datasets until a correct response – service error is returned
sDsN5	Continue to create non-persistent datasets until correct response-service error is returned
sDsN6	Create a persistent dataset with unknown members and verify response-service error
sDsN7	Create a non-persistent dataset with unknown members and verify response-service error
sDsN8	Delete a (pre-defined) non-deletable dataset, and verify response-service error
sDsN9	Delete a persistent dataset twice, and verify response-service error
sDsN10	Delete a non-persistent dataset twice, and verify response-service error
sDsN11	Delete a persistent dataset referenced by (report) control class, and verify response-service error (IEC 61850-7-2:2010, 13.1)
sDsN12	Delete a non- persistent dataset referenced by (report) control class, and verify response-service error (IEC 61850-7-2:2010, 13.1)
sDsN13	Request SetDataSetValues with a dataset with one or more read-only members, and verify response-service error

6.2.4.9 Substitution model – Positive test cases

The test cases listed in Table 9 shall apply.

Table 9 – Substitution positive test cases

Test case	Test case description
sSub1	Disable subEna and set subVal, subMag, subCMag, subQ, subID and verify the substituted values are not transmitted when subEna is disabled and are transmitted when subEna is enabled (IEC 61850-7-3:2010, Table 64).
sSub2	Verify that in case the association fails, the substituted values shall remain unchanged
sSub3	Verify setting subVal, subMag, subCMag, subQ and subID is allowed and the substituted values are transmitted and Quality.Source is set to Substituted when subEna is enabled

6.2.4.10 Unbuffered reporting model

6.2.4.10.1 Positive test cases

The test cases listed in Table 10 shall apply.

Table 10 – Unbuffered reporting positive test cases

Test case	Test case description
sRpt1	Request GetLogicalNodeDirectory(URCB) and check response Request GetURCBValues of all responded URCB's
sRpt2	Verify the reporting of optional fields of a URCB Configure/enable a URCB with all optional fields combinations: sequence-number, report-time-stamp, reason-for-inclusion, data-set-name, and/or data-reference (IEC 61850-7-2:2010, 17.2.3.2.2.1), force/trigger a report and check that the reports contain the enabled optional fields
sRpt3	Verify the trigger conditions of a URCB Configure and enable a URCB with optional fields: sequence-number, report-time-stamp, reason-for-inclusion, data-set-name, data-reference, buffer-overflow, and entryID and check that the reports are transmitted according to the following (supported) trigger conditions: • on integrity • on update (dupd) • on update with integrity • on data change (dchg) • on data and quality change • on data and quality change with integrity period Verify the validity of the ReasonCode (IEC 61850-7-2:2010,17.2.3.2.2.9) Verify that when more trigger conditions are met preferably only one report is generated (IEC 61850-7-2:2010,17.2.3.2.3.2) Verify that reports are only sent when RptEna is set to True (IEC 61850-7-2:2010,17.2.2.5), when reporting is disabled no reports should be transmitted
sRpt4	General interrogation (IEC 61850-7-2:2010,17.2.2.13) Setting the GI attribute of an URCB shall start the general-interrogation process. One report with the current data values will be sent. After initiation of the general-interrogation, the GI attribute is reset to False.
sRpt5	Segmentation of reports Verify that if a long report does not fit in one message, the report is split into sub-reports. Enable sequence-number and report-time-stamp optional field and check validity of: (IEC 61850-7-2:2010, 17.2.3.2.2.5) – SeqNum (not changed) – SubSeqNum (0 for first report, incrementing, roll-over) – MoreSegmentsFollow – TimeOfEntry (not changed as SeqNum is not altered) (IEC 61850-7-2:2010,17.2.3.2.2.9) Verify that an update of a data value during sending of a segmented report caused by an integrity or general-interrogation trigger can be interrupted by a report with change of one of the data values with a new sequence number (IEC 61850-7-2:2010,17.2.3.2.3.5) A new request for general-interrogation shall stop the sending of remaining segments of the GI-report that is still going on. A new GI-report shall start with a new sequence number and the sub-sequence number shall be 0 (IEC 61850-7-2:2010,17.2.3.2.3.4)
sRpt6	Configuration revision (IEC 61850-7-2:2010,17.2.2.7) Verify that ConfRev represents a count of the number of times the configuration of the data set referenced by DataSet has been changed. Changes that are counted are: • deletion of a member of the data-set • re-ordering of members in the data-set Verify that after a restart of the server, the value of ConfRev is restored to its original value of the base local configuration OR the value is retained from the configuration prior to restart (PIXIT) Verify that the server increments the ConfRev in case the data sets changes due to processing of ACSI services ConfRev should never be 0 (zero) in case DataSet is not null.
sRpt7	Verify that after a restart of the server, the value of ConRev is restored to its original value of the base local configuration OR the value is retained from the configuration prior to restart (PIXIT).

Test case	Test case description
sRpt8	Buffer Time (IEC 61850-7-2:2010,17.2.2.9) Verify that in the case where a second internal notification of the same member of a DATA-SET has occurred prior to the expiration of BufTm, the server: (IEC 61850-7-2:2010,17.2.2.9) • shall for status information behave as if BufTm has expired and immediately send the report, restart the timer with value BufTm and process the second notification or • may for analogue information behave as if BufTm has expired and immediately transmit the report for transmission, restart the timer with value BufTm and process the second notification or • may for analogue information substitute the current value in the pending report with the new one. Configure Buffer Time to 1 000 ms and force a data value change of multiple dataset members within buffer time. Server should send not more than one report per buffer time with all the data values changes since last report. Verify that the value 0 for buffer time indicates that the buffer time attribute is not used. (IEC 61850-7-2:2010,17.2.2.9) Verify that the BufTm value can contain at least the value 360 0000 (= 1 h in ms)
sRpt9	Verify that the DUT can send reports with data objects
sRpt10	Verify that the DUT can send reports with data attributes
sRpt11	Verify that the DUT can send any buffered events before the integrity report
sRpt12	Verify that when the LLN0 Behavior value is Off or On/Blocked, no data change/quality change reports should be transmitted anymore for the process values in this logical device (IEC 61850-7-4:2010, Annex A)
sRpt13	Verify that the server sets URCB Owner to a non-NULL value when the URCB is configured by a client and reset to NULL when a client releases the URCB
sRpt14	Verify that the DUT can process an URCB with maximum name length for RptID and Dataset (IEC 61850-7-2:2010, 22.2)

6.2.4.10.2 Negative test cases

The test cases listed in Table 11 shall apply.

Table 11 – Unbuffered reporting negative test cases

Test case	Test case description
sRpN1	Request GetURCBValues with wrong parameters and verify response- service error (IEC 61850-7-2:2010,17.2.5.3)
sRpN2	Configure reporting with trigger option GI (not dchg, qchg, dupd, integrity). When enabled, only GI reports are transmitted. No reports should be send when generating events (IEC 61850-7-2:2010,17.2.5.3)
sRpN3	Setting the integrity period to 0 with TrgOps = integrity will result in no integrity reports will be sent (IEC 61850-7-2:2010,17.2.2.12)
sRpN4	Incorrect configuration of a URCB: configure when enabled, configure ConfRev and SqNum and configure with unknown data set
sRpN5	Exclusive use of URCB and lost association Configure a URCB and set the Resv attribute and enable it. Verify another client cannot set any attribute of that URCB (IEC 61850-7-2:2010,17.2.4.5)
sRpN6	Configure unsupported URCB options (PIXIT); Configure unsupported trigger conditions, optional fields and related parameters
sRpN7	Verify that another client cannot configure a pre-assigned URCB
sRpN8	Verify that when TrgOps – GI is not set, the request to set GI to true shall fail

6.2.4.11 Buffered reporting model

6.2.4.11.1 Positive test cases

The test cases listed in Table 12 shall apply.

Table 12 – Buffered reporting positive test cases

Test case	Test case description
sBr1	Request GetLogicalNodeDirectory(BRCB) and check response Request GetBRCBValues of all responded BRCBs
sBr2	Verify the reporting of optional fields of a BRCB Configure/enable a BRCB with all optional fields combinations: sequence-number, report-time-stamp, reason-for-inclusion, data-set-name, data-reference, buffer-overflow, and/or entryID (IEC 61850-7-2:2010,17.2.3.2.2.1), force/trigger a report and check that the reports contain the enabled optional fields
sBr3	Verify the trigger conditions of a BRCB Configure and enable a BRCB with optional fields: sequence-number, report-time-stamp, reason-for-inclusion, data-set-name, data-reference, buffer-overflow, and entryID and check that the reports are transmitted according to the following (supported) trigger conditions: • on integrity • on update (dupd) • on update with integrity • on data change (dchg) • on data and quality change • on data and quality change with integrity period Verify the validity of the ReasonCode (IEC 61850-7-2:2010,17.2.3.2.2.9) Verify that when more trigger conditions are met preferably only one report is generated (IEC 61850-7-2:2010,17.2.3.2.3.2) Verify that reports are only sent when RptEna is set to True (IEC 61850-7-2:2010,17.2.2.5), when reporting is disabled no reports should be transmitted
sBr4	General interrogation (IEC 61850-7-2:2010,17.2.2.13) Setting the GI attribute of a BRCB shall start the general-interrogation process. One report with the current data values will be sent. After initiation of the general-interrogation, the GI attribute is reset to False.
sBr5	Segmentation of reports Verify that if a long report does not fit in one message, the report is split into sub-reports. Enable sequence-number and report-time-stamp optional field and check validity of: (IEC 61850-7-2:2010,17.2.3.2.2.5) – SqNum (not changed) – SubSqNum (0 for first report, incrementing, roll-over) – MoreSeqmentsFollow – TimeOfEntry (not changed as SqNum is not altered) (IEC 61850-7-2:2010,17.2.3.2.2.9) Verify that an update of a data value during sending of a segmented report caused by an integrity or general-interrogation trigger can be interrupted by a report with change of one of the data values with a new sequence number (IEC 61850-7-2:2010,17.2.3.2.3.5) A new request for general-interrogation shall stop the sending of remaining segments of the GI-report that is still going on. A new GI-report shall start with a new sequence number and the sub-sequence number shall be 0 (IEC 61850-7-2:2010,17.2.3.2.3.4) Verify that when OptFlds=sequence-number is NOT set, neither SubSqNum nor SqNum are present in the sub-reports (17.2.3.2.2.4, 17.2.3.2.2.5).

Test case	Test case description
sBr6	Configuration revision (IEC 61850-7-2:2010,17.2.2.7) Verify that ConfRev represents a count of the number of times the configuration of the data set referenced by DataSet has been changed. Changes that are counted are: • deletion of a member of the data-set • re-ordering of members in the data-set Verify that after a restart of the server, the value of ConfRev is restored to its original value of the base local configuration OR the value is retained from the configuration prior to restart (PICS) Verify that the server increments the ConfRev in case the data sets changes due to processing of ACSI services ConfRev should never be 0 (zero) in case DataSet is not null
sBr7	Verify that after a restart of the server, the value of ConRev is restored to its original value of the base local configuration OR the value is retained from the configuration prior to restart (PIXIT).
sBr8	Buffer Time (IEC 61850-7-2:2010,17.2.2.9) Verify that in the case where a second internal notification of the same member of a DATA-SET has occurred prior to the expiration of BufTm, the server: (IEC 61850-7-2:2010,17.2.2.9) • shall for status information behave as if BufTm has expired and immediately send the report, restart the timer with value BufTm and process the second notification or • may for analogue information behave as if BufTm has expired and immediately transmit the report for transmission, restart the timer with value BufTm and process the second notification or • may for analogue information substitute the current value in the pending report with the new one. Configure Buffer Time to 1 000 ms and force a data value change of multiple dataset members within buffer time. Server should send not more than one report per buffer time with all the data values changes since last report. Verify that the value 0 for buffer time indicates that the buffer time attribute is not used (IEC 61850-7-2:2010,17.2.2.9) Verify that the BufTm value can contain at least the value 360 0000 (= 1 h in ms)
sBr9	Verify that the DUT can send reports with data objects
sBr10	Verify that the DUT can send reports with data attributes
sBr11	Verify that all buffered events shall be sent before integrity reports can be sent (IEC 61850-7-2:2010,17.2.3.2.3.3)
sBr12	Verify that all buffered events shall be sent before the GI report can be sent (IEC 61850-7-2:2010,17.2.3.2.3.4)
sBr13	Verify that the server sets BRCB Owner to a non-NULL value when the BRCB is configured by a client and reset to NULL when a client releases the BRCB. For at pre-assigned BRCB the server resets the Owner to the pre-assigned client address.
sBr14	Verify that the DUT can process a BRCB with maximum name length for RptID and DataSet (IEC 61850-7-2:2010, 22.2)
sBr20	Buffered reporting (BRCB) state machine (IEC 61850-7-2:2010,17.2.2.5, Figure 24) – Verify that events are buffered after the association is released – Verify that reporting is disabled after the association is lost – Verify that not received reports while not associated are received now in the correct order (SOE) (IEC 61850-7-2:2010,17.2.1, IEC 61850-7-2:2010,17.2.2.5) – Do the same but now set PurgeBuf to True before enabling the reporting. No stored buffered reports should be send (IEC 61850-7-2:2010,17.2.2.14) – Verify that after changing DataSet, the report buffer is purged. (IEC 61850-7-2:2010,17.2.2.5) Force buffer overflow, the OptFlds buffer-overflow should be set in the first report that is sent with events that occurred after the overflow (IEC 61850-7-2:2010,17 2.3.2.2.8)
sBr21	Buffered reporting (BRCB); buffering events (IEC 61850-7-2:2010,17.2.3.2.3.6) Verify that after the association is available again and after the client has set the EntryID, and enabled the BRCB, the BRCB shall start sending the reports of events that have been buffered. The BRCB shall use the sequence and subsequence numbers so that no gaps occur.

Test case	Test case description
sBr22	Verify that integrity reports are buffered
sBr23	Verify successful ResvTms behavior - On ResvTms = –1 the BRCB can be used by the configured client - On ResvTms = 0 a client can reserve the BRCB by writing a value and configure the BRCB - On lost association the reserved BRCB is released after the ResvTms number of seconds (ResvTms set to zero) On lost association, within ResvTms time none of other clients can reserve the BRCB except the one who did it originally (the client restores association)
sBr24	Verify that a SetBRCBValues request, for setting ResvTms, shall: - Generate a negative response if the BRCB's ResvTms value is –1. - Generate a negative response if the BRCB's ResvTms value is non-zero and if the SetBRCBValues request is being issued by another client for whom the BRCB is not reserved. Generate a negative response if the ResvTms value to be set is negative.
sBr25	Verify that a change of one of the following BRCB parameters purges the buffer: RptID, BufTm, TrgOps, IntgPd, DatSet. A change of OptFlds should not purge the buffer (IEC 61850-7-2:2010,17.2.2.5)
sBr26	Verify that after setting an invalid, null or non-existing EntryID the DUT sends all reports in the buffer
sBr27	Verify that when the BRCB state is RptEna=FALSE a GetBRCBValues shall return the EntryID value that represents the last (newest) entry that has been entered into the buffer. And when the BRCB RptEna=TRUE: The value of EntryID, returned in a GetBRCBValues response, shall be the EntryID of the last EntryID formatted and queued for transmission.
sBr28	Verify that only the last buffered GI report is transmitted after a resync.

6.2.4.11.2 Negative test cases

The test cases listed in Table 13 shall apply.

Table 13 – Buffered reporting negative test cases

Test case	Test case description
sBrN1	Request GetBRCBValues with wrong parameters and verify response- service error (IEC 61850-7-2:2010,17.2.3.3.2)
sBrN2	Configure reporting with trigger option GI (not dchg, qchg, dupd, integrity). When enabled only GI reports are transmitted. No reports should be send when generating events (IEC 61850-7-2:2010,17.2.3.2.2.4)
sBrN3	Setting the integrity period to 0 with TrgOps = integrity will result in no integrity reports will be sent (IEC 61850-7-2:2010,17.2.2.11)
sBrN4	Incorrect configuration of a BRCB: configure when enabled, configure ConfRev and SqNum and configure with unknown data set
sBrN5	Exclusive use of BRCB and lost association Configure a BRCB and enable it. Verify another client cannot set attributes value in this BRCB. (IEC 61850-7-2:2010,17.2.1)
sBrN6	Configure unsupported BRCB options (PIXIT); Configure unsupported trigger conditions, optional fields and related parameters
sBrN7	Verify that another client cannot configure a pre-assigned BRCB
sBrN8	Verify that when TrgOps – GI is not set, the device does not send reports with reason code GI

6.2.4.12 Log model

6.2.4.12.1 Positive test cases

The test cases listed in Table 14 shall apply.

Table 14 – Log positive test cases

Test case	Test case description
sLog1	Request GetLogicalNodeDirectory(LOG) and check response+
sLog2	Request GetLogicalNodeDirectory(LCB) and check response+
sLog3	Request GetLCBValues with functional constraint LG of all responded LCBs
sLog4	Request SetLCBValues with functional constraint LG when LCB is disabled
sLog5	Verify that logging is independent of a limited set of external application associations or other communication transactions
sLog6	Configure and enable logging and check that the following logging trigger conditions place a correct entry in the log with the correct members of the data set – on integrity – on update (dupd) – on update with integrity – on data change (dchg) – on quality change (qchg) – on data and quality change – on data and quality change with integrity period
sLog7	Request QueryLogByTime and check response+
sLog8	Request QueryLogAfter and check response+
sLog9	Request GetLogStatusValues and check response+, verify that the responded entries indicate the oldest/newest entry ID/time available in the log
sLog10	Check that data is logged as defined in the settings of logical node GLOG. The corresponding reason code shall be "application-trigger"
sLog11	Verify that server can process a LCB and LOG with maximum name length for LCBRef, LogRef and DataSet (IEC 61850-7-2:2010, 22.2)
sLog12	Verify that log entries are volatile and not lost after reboot and power loss.

6.2.4.12.2 Negative test cases

The test cases listed in Table 15 shall apply.

Table 15 – Log negative test cases

Test case	Test case description
sLogN1	Request the following log services with wrong parameters (out of range entries, or non existent Dataset, LCB or Log) and verify response-service error – GetLCBValues (IEC 61850-7-2:2010,17.3.2.5) – SetLCBValues (IEC 61850-7-2:2010,17.3.2.6) – QueryLogByTime (IEC 61850-7-2:2010,17.3.5.2) – QueryLogAfter (IEC 61850-7-2:2010,17.3.5.3) – GetLogStatusValues (IEC 61850-7-2:2010,17.3.5.4)
sLogN2	Request SetLCBValues with functional constraint LG when LCB is enabled and verify response-service error

6.2.4.13 Control models

6.2.4.13.1 General

Testing of the control models has been divided in a general test and a list of test cases for the specific control models. The general test cases are as follows.

6.2.4.13.2 Control model general test cases

The test cases listed in Table 16 shall apply.

Table 16 – Control model test cases

Test case	Test case description
sCtl1	Force and check each path in control state machine for several control objects with control models • direct with normal security (IEC 61850-7-2:2010, 20.2.1) • SBO-control with normal security (IEC 61850-7-2:2010, 20.2.2) • direct with enhanced security (IEC 61850-7-2:2010, 20.3.2) • SBO-control with enhanced security (IEC 61850-7-2:2010, 20.3.3) Compare detailed test cases for each control model
sCtl2	Change control model using online services and verify that the control object responds according to the new control model
sCtl3	Time Operate a second enhanced security control object before the activation time of the first control object (PIXIT)
sCtl4	Verify that the stSeld attribute value is set/reset as specified in the state machines
sCtl5	Verify test flag in selectwithvalue/operate and Beh = test (IEC 61850-7-4:2010, Annex A ,Table A.1) • When LN Beh is "on", the control Requests are rejected with AddCause "Blocked-by-mode" • When LN Beh is "test/blocked", the control requests are accepted • When LN Beh is "test", the control requests are accepted
sCtl6	Select all SBO control objects and cancel them in opposite order. In case a control action is blocked because another control is already running, the AddCause shall be "1-of-n-control"
sCtl7	Verify that with interlock or synchrocheck conditions the specified checks are performed and the command is executed accordingly (IEC 61850-7-2:2010, 20.5.2.7) • When the interlock check fails with AddCause "Blocked-by-interlocking" • When the interlock check passes • When the synchro check fails with AddCause "Blocked-by-synchrocheck" • When the synchro check passes
sCtl8	Operate (without select) a SBO control object and verify that the request is rejected with AddCause "Object-not-selected" (IEC 61850-7-2:2010, Table 54)
sCtl9	Select the same control object twice, verify that the second select request is rejected with AddCause "Object-already-selected" (IEC 61850-7-2:2010, Table 54) and the object remains in selected state (Operate.req is accepted)
sCtl10	Operate control value is the same as the actual status value (On-On or Off-Off) and verify that the control request is rejected with AddCause "Position-reached" (IEC 61850-7-2:2010, Table 54, PIXIT)
sCtl11	Select the same control object from 2 different clients. Verify that the control requests from the second client are rejected with AddCause "Locked-by-other-client" (IEC 61850-7-2:2010, Table 54)
sCtl12	Select/Operate an unknown control object and verify that the control requests are rejected with AddCause "Unknown" (IEC 61850-7-2:2010, Table 54)
sCtl13	Verify that the Select request on a direct operate control object is rejected with Addcause "Unknown" (IEC 61850-7-2:2010, Table 54)
sCtl14	Operate the same direct control object twice from 2 clients (IEC 61850-7-2:2010, Table 54, PIXIT) and verify that the last control request is rejected with AddCause "Command-already-in-execution"
sCtl15	Verify that the SBOes Operate or Cancel request with different control parameters than the SelectWithValue is rejected with AddCause: Inconsistent-parameters"

Test case	Test case description
sCtl16	Verify that when Loc is set remote control requests are rejected with AddCause "Blocked-by-switching-hierarchy"
sCtl17	Verify that with plant level authority (LocSta=T) remote control requests are rejected with AddCause "Blocked-by-switching-hierarchy"
sCtl18	Verify that on CmdBlk.stVal is set the control requests are rejected with AddCause "Blocked-by-command" (IEC 61850-7-2:2010, Table 54)
sCtl19	Verify that when the blkEna is set the control requests are terminated/rejected with AddCause "Time-limit-over"
sCtl20	Verify that when parameters are changed after the select respond, the operate request is rejected with AddCause "Parameter-change-in-execution" (IEC 61850-7-2:2010, Table 54)
sCtl21	Verify that when tap changer has reached the limit (EndPosR or EndPosL in YLTC) control requests are rejected with AddCause "Step-limit" (IEC 61850-7-2:2010, Table 54)
sCtl22	Verify that with insufficient access authority control requests are rejected with AddCause "No-access-authority" (IEC 61850-7-2:2010, Table 54)
sCtl23	Verify that when a APC control action end position has overshoot the command terminates with AddCause "Ended-with-overshoot" (IEC 61850-7-2:2010, Table 47)
sCtl24	Verify that when a APC control action is aborted due to deviation between the command value and the measured value the control terminates with AddCause "Abortion-due-to-deviation" (IEC 61850-7-2:2010, Table 47)
sCtl25	Verify that a cancel request is successful when the control object is in the unselected state (IEC 61850-7-2:2010, Table 47)
sCtl26	Verify that when the control object is in the WaitForExecution state the cancel or SelectWithValue request is rejected with AddCause "Command-already-in-execution" (IEC 61850-7-2:2010, Table 54)
sCtl27	Verify that the SelectWithValue request on a SBOs control object is rejected with Addcause "Unknown" (IEC 61850-7-2:2010, Table 54)
sCtl28	Verify that the DUT can control an object with maximum name length for IED and Logical Device (IEC 61850-7-2:2010, 22.2)

The testing of the specific control models has been divided into the four possible control models that can be implemented:

- a) Direct control with normal security (DOs),
- b) SBO control with normal security (SBOs),
- c) Direct control with enhanced security (DOes),
- d) SBO control with enhanced security (SBOes).

6.2.4.13.3 Control model specific test cases

The test cases listed in Table 17, Table 18, Table 19 and Table 20 shall apply.

Table 17 – DOns test cases

Test case	Test case description
sDOns1	Send a correct Operate request
sDOns2	Send an Operate request, resulting in "Test not ok"
sDOns3	Send an TimeActivatedOperate request, resulting in respond-
sDOns4	Send a correct TimeActivatedOperate request Verify the TimeActivatedOperateTermination+
sDOns5	Send a TimeActivatedOperate request, Verify that each of these paths will return the device to the Ready state and verify the TimeActivatedOperateTermination-: – Force a "Test not ok" – Send a correct Cancel request

Table 18 – SBOns test cases

Test case	Test case description
sSBOns1	Send a correct Select request Send a correct Operate request
sSBOns2	Send a correct Select request Verify that each of these paths will return the device to the Unselected state: – Send a correct Cancel request – Wait for timeout – Association lost – Send a Release request – Send an Operate request resulting in "Test not ok"
sSBOns3	Send a correct Select request – Send an incorrect TimeActivatedOperate request resulting in respond-
sSBOns4	Send a correct Select request Send a correct TimeActivatedOperate request, thereby making sure the device will generate a 'test Ok'. Verify the TimeActivatedOperateTermination+
sSBOns5	Send a correct Select request Send a correct TimeActivatedOperate request. Verify that each of these paths will return the device to Ready state and the TimeActivatedOperateTermination-: – Force a test "Test not ok" – Send a correct Cancel request
sSBOns6	Send a correct Select request resulting in respond- Verify that each of the device returns to the Unselected state:
sSBOns7	Send a correct Select request Verify that sending multiple Operate Many requests will return the device to Ready state Verify that sending a Cancel request will return the device to the Unselected state

Table 19 – DOes test cases

Test case	Test case description
sDOes1	Send a correct Operate request Verify that each of these paths will return the device to the Ready state and verify the CommandTermination: – force the equipment simulator to move to the requested new state – force the equipment simulator to keep the old state (AddCause: Time-limit-over or Invalid-position) – force the equipment simulator to move to the 'between' state (AddCause: Invalid-position)
sDOes2	Send an Operate request, thereby making sure the device will generate a 'Test not Ok'.
sDOes3	Send a TimeActivated Operate request, resulting in respond-
sDOes4	Send a correct TimeActivated Operate request Verify the TimeActivatedOperateTermination+ Verify that each of these paths will return the device to the Ready state and verify the CommandTermination: – force the equipment simulator to move to the requested new state – force the equipment simulator to keep the old state (AddCause: Time-limit-over) – force the equipment simulator to move to the 'between' state (AddCause: Invalid-position)
sDOes5	Send a correct TimeActivated Operate request Verify the TimeActivatedOperateTermination+ Verify that each of these paths will return the device to the Ready state and verify the TimeActivatedOperateTermination: – Force a “Test not ok” – Send a correct Cancel request

Table 20 – SBOes test cases

Test case	Test case description
sSBOes1	Send a correct SelectWithValue and Operate request Verify that each of these paths will return the device to the Unselected state Verify the command termination: – force the equipment simulator to move to the requested new state – force the equipment simulator to keep the old state (AddCause: Time-limit-over or Invalid-position) – force the equipment simulator to move to the 'between' state (AddCause: Invalid-position) –
sSBOes2	Send a correct SelectWithValue request Verify that each of these paths will return the device to the Unselected state: – Send a correct Cancel request – Client waits for select timeout – Send a Release request – Send an Operate request resulting in "Test not ok"
sSBOes3	Send a correct SelectWithValue and TimeActivatedOperate request, resulting in respond-
sSBOes4	Send a correct SelectWithValue request Send a correct TimeActivatedOperate Once request Verify the TimeActivatedOperateTermination+ Verify that each of these paths will return the device to the Unselected state and verify the CommandTermination: – force the equipment simulator to move to the requested new state – force the equipment simulator to keep the old state (AddCause:Time-limit-over or Invalid-position) – force the equipment simulator to move to the 'between' state (AddCause: Invalid-position)
sSBOes5	Send a correct SelectWithValue request Send a correct TimeActivatedOperate request Verify that each of these paths will return the device to the Ready state: – Force a test "Test not ok" – Send a correct Cancel request
sSBOes6	Select device using SelectWithValue with improper access rights. Access should be denied (IEC 61850-7-2:2010,20.2.2) or send an incorrect SelectWithValue request
sSBOes7	Send a correct SelectWithValue request Verify that sending multiple Operate Many requests will return the device to Ready state Verify that sending a Cancel request will return the device to the Unselected state
sSBOes8	Verify that the Operate or Cancel request with different control parameters than the SelectWithValue is rejected with AddCause: Inconsistent-parameters

6.2.4.14 Time and time synchronisation model

6.2.4.14.1 General

The time and time synchronization test cases defined for a server device are as follows.

6.2.4.14.2 Positive test cases

The test cases listed in Table 21 shall apply.

Table 21 – Time positive test cases

Test case	Test case description
sTm1	Verify that the DUT supports and executes the SCSM time synchronisation
sTm2	Check report/logging timestamp accuracy matches the documented timestamp quality of the server
sTm3	Verify that when the device supports time zones and daylight saving the time stamp of events and disturbance files are UTC time
sTm4	Verify the time management settings in logical node LTIM
sTm5	Verify the time master supervision in logical node LTMS

6.2.4.14.3 Negative test cases

The test case listed in Table 22 shall apply.

Table 22 – Time negative test cases

Test case	Test case description
sTmN1	Verify that when time synchronisation communication lost is detected after a time period specified by the test facility.
sTmN2	On synchronisation error, deviation beyond time stamp tolerance should be detected

6.2.5 Test cases to test a client device

6.2.5.1 General

This part of the IEC 61400-25-5 specifies abstract test cases for testing the compliance of a IEC 61400-25 client. The abstract test cases shall be used for the definition of test procedures to run in tests.

The difference to the specifications given in the IEC 61850-10 is the wind power information model and the wind power specific mappings. The specific IEC 61400-25-2 data classes are not tested. The IEC 61400-25-4 mapping particularities are not described in the IEC 61400-25-5 abstract level.

The SCSM specific test procedures shall be provided by test facilities agreed upon by the market participants

6.2.5.2 Test system architecture to test a client device

In order to be able to perform a client device test, a minimum test set-up is necessary. The test architecture contains (see Figure 5):

- DUT with optional HMI;
- multi server simulator to respond to TPAA messages from the DUT;
- test master to start/stop test cases, start/stop the analyse and archive test results;
- time master;
- engineering tool to configure the DUT;
- a protocol analyser to store all the network traffic for each test case.

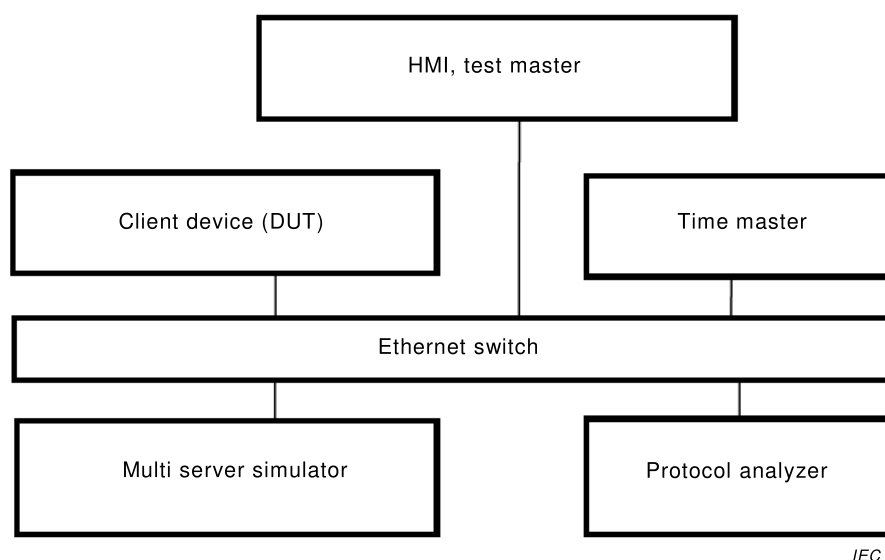


Figure 5 – Test system architecture to test a client device

The test system shall include documentation regarding test system hardware and test system software.

6.2.5.3 Documentation and version control test procedure overview

The test case listed in Table 23 shall apply.

Table 23 – Client documentation test case

Test case	Test case description
cDoc1	Check if the major/minor software version in the PICS documentation and the DUT do match (IEC 61850-4)
cDoc2	Check if the major/minor software version manufacturer PIXIT documentation and software version of the DUT do match (IEC 61850-4). PIXIT shall indicate the required information as requested in the test cases
cDoc3	Check if the major/minor software version manufacturer MICS documentation and software version of the DUT do match (IEC 61850-4). MICS shall indicate which CDCs and/or CDC parts are supported by the DUT, for example arrays
cDoc4	Check if the major/minor software version in the TICS documentation and the software version of the DUT do match (IEC 61850-4). TICS shall indicate the supported technical issues.

6.2.5.4 Data model test cases

The test case listed in Table 24 shall apply.

Table 24 – Client data model test case

Test case	Test case description
cMdl1	Verify that the client can handle the maximum name length according to IEC 61400-25-2 and SCSM expands objects like SDOs correctly (PIXIT)
cMdl2	Verify that DUT supports the following naming conventions for the supported control blocks a) unbuffered report control block – not indexed b) unbuffered report control block – indexed c) buffered report control blocks d) Log control block
cMdl3	Verify that DUT can read and process the mandatory and optional attributes from the CDCs in IEC 61400-25 unless stated otherwise in the MICS

6.2.5.5 Mapping of ACSI models and services test cases

Test items shall be grouped together in tables. The tables shall reflect the applicable service models specified in IEC 61400-25-3:

- Application association (cAss);
- Server, Logical device, Logical node, Data, and Data Attribute model (cSrv);
- Data set model (cDs);
- Unbuffered report control model (cRp);
- Buffered report control model (cBr);
- Log control model (cLog);
- Control model (cCtl);
- Time and time synchronisation model (cTm).

Test cases are defined for each ACSI model and services in the following categories:

- positive = verification of normal conditions, typically resulting in response+;
- negative = verification of abnormal conditions, typically resulting in response–.

A test case is mandatory when the applicable ACSI model and ACSI service is supported by the DUT. This is specified in the PICS according to IEC 61850-7-2:2010, Annex A. The test result interpretation (passed/failed) depends on the declared capabilities.

6.2.5.6 Application association model

6.2.5.6.1 Positive test case

The test cases listed in Table 25 shall apply.

Table 25 – Association positive test cases

Test case	Test case description
cAss1	Associate and force the DUT to release or abort a TPAA association (IEC 61850-7-2:2010,8.3)
cAss2	Force the DUT to associate with maximum number of servers simultaneously (PIXIT)
cAss3	Verify that the DUT restores the association after the association of one server is lost and that this has no effect on the other active associations of the other servers

6.2.5.6.2 Negative test case

The test cases listed in Table 26 shall apply.

Table 26 – Association negative test cases

Test case	Test case description
cAssN1	Associate and server responds with negative answer due to AccessPointReference.
cAssN2	Associate and server responds with negative answer due to AuthenticationParameter.
cAssN3	Associate and server releases TPAA association (IEC 61850-7-2:2010,8.3). DUT should try to re-establish the association after the configured period (PIXIT).
cAssN4	Associate and server-abort TPAA association (IEC 61850-7-2:2010,8.3). DUT should try to re-establish the association after the configured period (PIXIT).
cAssN5	Associate and server denies TPAA association (IEC 61850-7-2:2010,8.3) DUT should try to re-establish the association after the configured period (PIXIT).
cAssN6	Disconnect the communication interface between server and the Ethernet switch such that the link between DUT and the Ethernet switch stays active. The DUT shall detect link lost within a specified period. Once the link is re-established the DUT should try to establish the association again.
cAssN7	Interrupt and restore the power supply, the DUT shall establish the configured associations when ready (PIXIT).

6.2.5.7 Server, Logical Device, Logical Node, and Data model

6.2.5.7.1 Positive test cases

The test cases listed in Table 27 shall apply.

Table 27 – Server positive test cases

Test case	Test case description
cSrv1	If the DUT implements Autodescription (See Note 1), force the DUT to start the autodescription and check that the DUT requests a GetServerDirectory (LOGICAL-DEVICE) to all the logical devices of the configured servers (See Note 2) (IEC 61850-7-2:2010,7.2.2)
cSrv2	If DUT implements Autodescription, for each GetServerDirectory (LOGICAL-DEVICE) response, check that the DUT issues a GetLogicalDeviceDirectory request (IEC 61850-7-2:2010,9.2.1)
cSrv3	If DUT “implements Autodescription”, for each GetLogicalDeviceDirectory response issue, check that the DUT issues a GetLogicalNodeDirectory(DATA) request (IEC 61850-7-2:2010,10.2.2)
cSrv4	If the DUT “implements Autodescription”, for a subset of the GetLogicalNodeDirectory(DATA) response, check that the DUT issues at least one of the following services: – GetDataDirectory request and check response (IEC 61850-7-2:2010,11.4.4), – GetDataDefinition request and check response (IEC 61850-7-2:2010,11.4.5).
cSrv5	Verify that after startup, the DUT is able to update the process values of the configured servers (PIXIT).
cSrv6	Request a SetDataValues of the different basic types (with e.g. FC=CF) and check the services (IEC 61400-25-3).
cSrv7	Request a GetDataValues and check if the DUT updates its model
cSrv8	Request GetAllDataValues for the required functional constraint and check if the DUT updates its model (IEC 61850-7-2:2010,10.2.3).
cSrv9	Verify that the DUT is able to set/reset blkEna (IEC 61850-7-3:2010,6.2.6)
NOTE 1 Implement Autodescription means that there is a way to configure the client to update the image of the model of one of the servers it has to communicate with using the ACSI services.	
NOTE 2 Configured servers means the servers that the DUT is configured to communicated with. The DUT at least needs to know the parameters to establish an association with them.	

6.2.5.7.2 Negative test cases

The test cases listed in Table 28 shall apply.

Table 28 – Server negative test cases

Test case	Test case description
cSrvN1	If the DUT implements Autodescription, force the SUT to start the autodescription and check that the DUT still communicates with other servers when it requests the following services with negative response: a) GetServerDirectory(LOGICAL-DEVICE), b) GetLogicalDeviceDirectory, c) GetLogicalNodeDirectory(DATA), d) GetDataDirectory, e) GetDataDefinition.
cSrvN2	Check that the DUT is able to communicate with other connected servers after a request for GetAllDataValues fails in the following circumstances: a) The response is negative. b) The response comes with mismatching data objects.
cSrvN3	Check that the DUT is able to communicate with other connected servers after a request for GetDataValues fails in the following circumstances: a) The response is negative. b) The response comes with mismatching data objects. c) The value is out of the valid range for this data.
cSrvN4	Check that the DUT is able to communicate with other connected servers after a request for SetDataValues fails in the following circumstances: a) The response is negative. b) One of the data values is read-only
cSrvN5	If the DUT detects/notify changes in the “Quality” attribute, use the SERVER SIMULATOR to force different values in the Quality of the measured/status values monitored by the DUT and check the behaviour described in the PIXIT.
cSrvN6	If the DUT detects/notify changes in the timeStamp’s “TimeQuality” attribute, use the SERVER SIMULATOR to force different values in the TimeQuality of the measured/status values monitored by the DUT and check the behaviour described in the PIXIT.

6.2.5.8 DataSet model

6.2.5.8.1 Positive test cases

The test cases listed in Table 29 shall apply.

Table 29 – Data set positive test cases

Test case	Test case description
cDs1	If the DUT implements Autodescription, force it to start Autodescription and check if it requests a GetLogicalNodeDirectory(DATASET) of the Logical Nodes of the configured servers (IEC 61850-7-2:2010,10.2.2)
cDs2	If the DUT implements Autodescription, force it to start autodescription and check it requests a GetDataSetDirectory of all the DataSets of the server (IEC 61850-7-2:2010,13.3.6)
cDs3	Check that the DUT can request a GetDataSetValues and handle the respond (IEC 61850-7-2:2010,13.3.2)
cDs4	Check that the DUT can request a SetDataSetValues and handle the respond (IEC 61850-7-2:2010, 13.3.3)
cDs5	Verify that the DUT checks the pre-configured datasets in the SCD file. If any deviation is detected the DUT behaves as specified in the PIXIT
cDs6	If the DUT creates persistent / non-persistent datasets dynamically after starting up check that the DUT sends the CreateDataSet services according to configuration. PIXIT (IEC 61850-7-2:2010,13.3.4)
cDs7	Request a DeleteDataSet service and check that the DUT sends the request properly and is able to process the response of the server (IEC 61850-7-2:2010,13.3.5)
cDs8	Verify that a persistent data set can be handled with the maximum name length for data set and a data set member (IEC 61850-7-2:2010, 22.2)
cDs9	Verify that a non-persistent data set can be handled with the maximum name length for data set and a data set member (IEC 61850-7-2:2010, 22.2)

6.2.5.8.2 Negative test cases

The test cases listed in Table 30 shall apply.

Table 30 – Data set negative test cases

Test case	Test case description
cDsN1	If the DUT implements Autodescription, force the DUT to start the Autodescription and check that the DUT still communicates with other servers when it request the following services with negative response: a) GetLogicalNodeDirectory (DATA-SET), b) GetDataSetDirectory.
cDsN2	Check that the DUT still communicates with other servers properly when it requests a GetDataSetValue to one of them and the following situations happen: a) the response is negative, b) the response comes with more/less elements than expected, c) the response comes with reordered elements of different types, d) the response comes with reordered elements of the same type.
cDsN3	Check that the DUT still communicates with other servers properly when it requests a SetDataSetValue to one of them and the response is negative.
cDsN4	If the DUT creates persistent/non-persistent datasets dynamically after starting up check that the DUT still communicates with other servers when it requests a CreateDataSet with negative response
cDsN5	If the DUT configures the datasets dynamically after starting up check that the DUT still communicates with other servers when it requests a DeleteDataSet with negative response

6.2.5.9 Substitution model – Test cases

The test cases listed in Table 31 shall apply.

Table 31 – Substitution test cases

Test case	Test case description
cSub1	Verify that the DUT can enable substitution, enter a substituted value and disable substitution
cSub2	Verify that the DUT can display the source “substituted” for substituted values
cSub3	Verify that the DUT can display the source “substituted” for values substituted by another client
cSub4	Verify that the DUT can handle the maximum name length for substitution values (IEC 61850-7-2:2010,22.2)

6.2.5.10 Unbuffered reporting model

6.2.5.10.1 Positive test cases

The test cases listed in Table 32 shall apply.

Table 32 – Unbuffered reporting positive test cases

Test case	Test case description
cRp1	If the DUT implements Autodescription, force it to start Autodescription and check if it requests a GetLogicalNodeDirectory(URCB) of the logical nodes declared in the PIXIT of all configured servers
cRp2	If the DUT configures the server's Unbuffered ReportControlBlock parameters after startup using SetURCBValues, check that the SetURCBValues are sent with the configured values (IEC 61850-7-2:2010,17.2.5.4)
cRp3	Verify that the DUT is able to process the reports with different optional fields: Force the DUT to configure/enable a URCB with useful optional fields combinations: sequence-number, report-time-stamp, reason-for-inclusion, data-set-name and/or data-reference (IEC 61850-7-2:2010,17.2.2.4), force/trigger a report and check that the DUT is able to process the reports and updates its database.
cRp4	Verify that the DUT is able to process the reports with different trigger conditions (IEC 61850-7-2:2010,17.2.2.4): Configure and enable a URCB with all supported optional fields and check that the reports are transmitted according to the following (supported) trigger conditions: a) on integrity b) on update (dupd) c) on update with integrity d) on data change (dchg) e) on data change and quality change (dchg, qchg) f) on data change and quality change with integrity period (dchg, qchg)
cRp5	Verify that the DUT is able to process segmented reports
cRp6	Verify that the DUT can change the (pre-) configured Buffer Time (IEC 61850-7-2:2010,17.2.2.9)
cRp7	Verify that the DUT can force a General interrogation (IEC 61850-7-2:2010,17.2.2.13)
cRp8	Verify that after startup the DUT configures and enables the URCB's as specified in the SCD file
cRp9	Verify that the DUT can handle reporting of complex structured data (for example WYE and DEL data objects)
cRp10	Verify that the DUT can handle reporting of basic data (for example stVal and quality)
cRp11	Verify that the DUT can handle a URCB, RptID and DataSet with maximum name length (IEC 61850-7-2:2010,22.2)
cRp12	Verify that the DUT can change the dataset elements of a dynamic dataset previously used in a URCB resulting in a ConfRev increment by the server
cRp13	Verify that the DUT configures another indexed URCB when another DUT has reserved the indexed URCB before

6.2.5.10.2 Negative test cases

The test cases listed in Table 33 shall apply.

Table 33 – Unbuffered reporting negative test cases

Test case	Test case description
cRpN1	If the DUT implements Autodescription, force the DUT to start the Autodescription and check that the DUT still communicates with other servers when it requests GetLogicalNodeDirectory (URCB) with negative response.
cRpN2	Check that the DUT still works properly when it requests a GetURCBValues when the response is negative.
cRpN3	Check that the DUT still works properly when it requests a SetURCBValues when the response is negative.
cRpN4	Check that the DUT still works properly when it requests a SetURCBValues and the URCB is reserved (Resv=TRUE, PIXIT).
cRpN5	Report with not supported OptFlds. Check that the DUT does not collapse if it receives a Report with a non-configured or non-supported OptFlds.
cRpN6	Report with not supported TrgOps. Check that the DUT does not collapse if it receives a Report with a non-configured or non-supported Trigger Option.
cRpN7	Mismatching reports: a) Report with unknown DataSet b) Report with unknown RptId c) Report with incorrect references of the Data d) Report with incorrect types in the Data Check the behaviour described in the PIXIT.
cRpN8	Verify that the DUT detects a change in the ConfRev attribute (Configuration revision, IEC 61850-7-2:2010,14.2.2.7) of the Report Control Block. When the DUT does not perform the ConfRev check it should check the dataset elements. The means of detection needs to be specified in the PIXIT.

6.2.5.11 Buffered reporting model

6.2.5.11.1 Positive test cases

The test cases listed in Table 34 shall apply.

Table 34 – Buffered reporting positive test cases

Test case	Test case description
cBr1	If the DUT implements Autodescription, force it to start Autodescription and check if it requests a GetLogicalNodeDirectory(BRCB) of the logical nodes declared in the PIXIT of all configured servers
cBr2	If the DUT configures the server's Buffered ReportControlBlock parameters after startup using SetBRCBValues, check that the GetBRCBValues/SetBRCBValues are sent with the configured values (IEC 61850-7-2:2010,17.2.3.4)
cBr3	Verify that the DUT is able to process the reports with different optional fields: Force the DUT to configure/enable a BRCB with the useful optional fields combinations: sequence-number, report-time-stamp, reason-for-inclusion, data-set-name, data-reference, buffer-overflow, entryID and conf-version, force/trigger a report and check that the DUT is able to process the reports and updates its database (IEC 61850-7-2:2010,17.2.2.8)
cBr4	Verify that the DUT is able to process the reports with different trigger conditions (IEC 61850-7-2:2010,17.2.2.11) Configure and enable a BRCB with all useful optional fields: sequence-number, report-time-stamp, reason-for-inclusion, data-set-name, data-reference, buffer-overflow, entryID and conf-revision and check that the reports are transmitted according to the following (supported) trigger conditions: a) on integrity b) on update (dupd) c) on update with integrity d) on data change (dchg) e) on data and quality change (dchg, qch) f) on data and quality change with integrity period (dchg, qchg)
cBr5	Verify that the DUT is able to process segmented reports
cBr6	Verify that the DUT can change the (pre-) configured Buffer Time (IEC 61850-7-2:2010,17.2.2.9)
cBr7	Verify that the DUT can force a General interrogation (IEC 61850-7-2:2010,17.2.2.13)
cBr8	Verify that after startup the DUT configures and enables the BRCBs as configured in the SCD file (and actually used).
cBr9	Verify that the DUT can handle reporting of complex structured data (for example WYE and DEL data objects)
cBr10	Verify that the DUT can handle reporting of basic data (for example stVal and quality)
cBr11	Verify that the DUT can handle a BRCB, RptID and DataSet with maximum name length (IEC 61850-7-2:2010,22.2)
cBr12	Verify that the DUT can change the dataset elements of a dynamic dataset previously used in a BRCB resulting in a ConfRev increment by the server
cBr13	Verify that the DUT configures another indexed BRCB when another client has reserved the indexed BRCB before.
cBr14	Verify that the DUT is able to process reports buffered during an lost association a) without bufferoverflow (PIXIT) b) with bufferoverflow
cBr15	Verify that the DUT is able to request specific buffered reports after restoring a lost association by setting the EntryID
cBr16	Verify that the DUT is able to purge buffered reports
cBr17	Verify that the DUT first sets the ResvTms attribute if this attribute is available and has value 0

6.2.5.11.2 Negative test cases

The test cases listed in Table 35 shall apply.

Table 35 – Buffered reporting negative test cases

Test case	Test case description
cBrN1	If the DUT implements Autodescription, force the DUT to start the Autodescription and check that the DUT still communicates with other servers when it requests GetLogicalNodeDirectory (BRCB) with negative response.
cBrN2	Check that the DUT still works properly when it requests a GetBRCBValues when the response is negative.
cBrN3	Check that the DUT still works properly when it requests a SetBRCBValues when the response is negative.
cBrN4	Check that the DUT still works properly when it requests a SetBRCBValues and the BRCB is used by or pre-assigned to another DUT (PIXIT).
cBrN5	Report with not supported OptFlds. Check that the DUT does not collapse if it receives a Report with a non-configured or non-supported OptFlds.
cBrN6	Report with not supported TrgOps. Check that the DUT does not collapse if it receives a Report with a non-configured or non-supported Trigger Option.
cBrN7	Mismatching reports: a) Report with unknown DataSet. b) Report with unknown RptID c) Report with incorrect references of the Data (when data references are enabled) d) Report with incorrect types in the Data Check the behaviour described in the PIXIT.
cBrN8	Verify that the DUT detects a change in the ConfRev attribute (Configuration revision, IEC 61850-7-2:2010, 17.2.2.7) of the Report Control Block. When the DUT does not perform the ConfRev check it should check the dataset elements. The means of detection needs to be specified in the PIXIT.
cBrN9	Verify that the DUT can handle a severe buffer overflow with SetBRBValues(EntryID) response-

6.2.5.12 Log model

6.2.5.12.1 Positive test cases

The test cases listed in Table 36 shall apply.

Table 36 – Log positive test cases

Test case	Test case description
cLog1	If the DUT implements Autodescription, force it to start Autodescription and check if it requests a GetLogicalNodeDirectory (LOG) of all the logical nodes of all configured servers
cLog2	If the DUT implements Autodescription, force it to start Autodescription and check if it requests a GetLogicalNodeDirectory(LCB) of all the logical nodes of all configured servers
cLog3	If the DUT implements Autodescription, force it to start Autodescription and check if it requests a GetLogStatusValues of all the LOGs found with the GetLogicalNodeDirectory(LCB) services
cLog4	If the DUT implements Autodescription, force it to start Autodescription and check if it requests a GetLCBValues of all the LCBs found with the GetLogicalNodeDirectory(LCB) services (IEC 61850-7-2:2010,17.3.2.5)
cLog5	If the DUT configures the server's LogControlBlock parameters after startup using SetLCBValues, check that the GetLCBValues/SetLCBValues are sent with the configured values (IEC 61850-7-2:2010,17.3.2.6)
cLog6	Force the DUT to enable the Logging of at least one LOG of the server and check that the DUT sent the request correctly
cLog7	Force the DUT to QueryLogByTime and check that the DUT updates its database with the Log entries received (IEC 61850-7-2:2010,17.3.5)
cLog8	Verify that the DUT can handle a LCB and DatSet with maximum name length (IEC 61850-7-2:2010,22.2)

6.2.5.12.2 Negative test cases

The test cases listed in Table 37 shall apply.

Table 37 – Log negative test cases

Test case	Test case description
cLogN1	If the DUT implements Autodescription, force the DUT to start the Autodescription and check that the DUT still communicates with other servers when it requests GetLogicalNodeDirectory (LCB) and GetLogicalNodeDirectory (LOG) with negative response.
cLogN2	Check that the DUT still works properly when it requests a GetLCBValues/GetLogStatusValues when the response is negative.
cLogN3	Check that the DUT still works properly when it requests a SetLCBValues when the response is negative.

6.2.5.13 Control model**6.2.5.13.1 General**

Testing of the control models has been divided in a list of general test cases and a list of test cases for the specific control models. The general test cases are as follows.

6.2.5.13.2 Positive test cases

The test cases listed in Table 38 shall apply.

Table 38 – Control model positive test cases

Test case	Test case description
cCtl1	Check if the DUT is able to set the TEST field in the SelectWithValue and Operate requests (PIXIT).
cCtl2	Check if the DUT is able to set the CHECK (Synchro-Check or Interlock-Check bits) in the commands (PIXIT) for the supported control models.
cCtl3	Check if the DUT is able to change control model using online services (PIXIT).
cCtl4	Verify the values of originator category and identification and the control number values (PIXIT).

6.2.5.13.3 Negative test cases

The test cases listed in Table 39 shall apply.

Table 39 – Control model negative test cases

Test case	Test case description
cCtlN1	Check if the DUT reacts in a proper way when its detects a control model mismatch (PIXIT): a) Server status-only – DUT expects controllable b) Server SBO – DUT expects direct operate c) Server direct operate – DUT expects

6.2.5.13.4 Control model specific test cases

The testing of the specific control models has been divided into the four possible control models that can be implemented:

- a) SBO control with enhanced security (SBOes),
- b) SBO control with normal security (SBOs),
- c) Direct control with enhanced security (DOes),
- d) Direct control with normal security(DOs),

The test cases listed in Table 40, Table 41, Table 42 and Table 43 shall apply.

Table 40 – SBOes test cases

Test case	Test case description
cSBOes1	SelectWithValue [test not ok] resp-: Select device using SelectWithValue resulting in test not ok. Check that the DUT indicates an error.
cSBOes2	SelectWithValue [test ok] resp+ and OperReq[test ok] resp+ of selected object Select device using correct SelectWithValue. Perform a correct Operate request. Check that the DUT indicates no error after receiving the command termination+
cSBOes3	SelectWithValue [test ok] resp+ and OperReq[test not ok] resp- of selected object. Perform a SelectWithValues and Operate request. The Operate results in test not ok. Check that the DUT realizes the operation failed.
cSBOes4	SelectWithValue [test ok] resp+ and CancelReq of selected object. Perform a correct Cancel request. Check that the DUT indicates no error.
cSBOes5	SelectWithValue [test ok] resp+ and TimOperReq[test ok] resp+ of selected object Perform a correct TimOperate request. Check that the DUT realizes the operation succeeded after the WaitForActionTime and detects the CommandTermination with the result of the order.
cSBOes6	SelectWithValue [test ok] resp+ and TimOperReq[test ok] resp- of selected object Perform a SelectWithValue and TimOperate request. The TimeOperate results in test not ok. Check that the DUT realizes the operation failed.

Table 41 – SBOs test cases

Test case	Test case description
cSBOs1	SelectReq[test not ok] resp-: DUT requests Select resulting in Test not ok. Check that the DUT realizes the select failed (PIXIT).
cSBOs2	SelectReq[test ok] resp+ and OperReq[test ok] resp+ of selected object. Select a controllable object using Select. Perform a correct Operate request. Check that the DUT does not generate an error.
cSBOs3	SelectReq[test ok] resp+ and OperReq[test not ok] resp- of selected object. Perform a correct Operate request resulting in Test not ok. Check that the DUT realizes the operation failed.
cSBOs4	SelectReq[test ok] resp+ and CancelReq of selected object. Perform a correct cancel request.
cSBOs5	SelectReq[test ok] resp+ and TimOperReq[test ok] resp+ of selected object. Perform a correct TimOperate request. Check that the DUT realizes the operation succeeded after the WaitForActionTime.
cSBOs6	SelectReq[test ok] resp+ and TimOperReq[test not ok] resp- of selected object. Perform a correct TimOperate request resulting in test not ok. Check that the DUT realizes the operation failed.

Table 42 – DOes test cases

Test case	Test case description
cDOes1	OperReq[test ok] resp+: Send a correct Operate request. a) Check that the DUT notice the operation ended positively when it receives the CommandTermination+. b) Check that the DUT notice the operation ended negatively when it receives the CommandTermination- (PIXIT)
cDOes2	OperReq[test not ok] resp-: Send an Operate request, thereby making sure the device will generate a 'test not Ok'. Check that the DUT realizes the operation failed (PIXIT)
cDOes3	TimOperReq[test not ok] resp-: Send a TimeActivated Operate request, thereby making sure the device will generate a 'test not Ok'. Check that the DUT realizes the operation failed.
cDOes4	TimOperReq[test ok] resp+: Send a correct TimeActivated Operate request. a) Check that the DUT realizes the operation request succeeded. b) Check that the DUT notice the operation ended positively when it receives the CommandTermination+. c) Check that the DUT notice the operation ended negatively when it receives the CommandTermination-.

Table 43 – DOns test cases

Test case	Test case description
cDOns1	OperReq[test ok] resp+ Perform a correct Operate request. Check that the DUT does not generate an error.
cDOns2	OperReq[test not ok] resp- DUT requests Oper resulting in Test not ok. Check that the DUT realizes the operation failed.
cDOns3	TimOperReq[test not ok] resp- DUT requests TimOper resulting in Test not ok. Check that the DUT realizes the time operation failed.
cDOns4	TimOperReq[test ok] + TimerExpired[test ok] resp+ Send a TimeActivatedOperate request, thereby making sure the device will generate a 'test Ok'. Verify that the WaitForActionTime results in a timer expired 'Test ok' and that the DUT realizes the operation succeeded.
cDOns5	TimOperReq[test ok] + TimerExpired[test not ok] resp- Send a TimeActivatedOperate request, thereby making sure the device will generate a 'test Ok'. Force situation that the WaitForActionTime results in a timer expired 'Test not ok'. Check that the DUT realizes the operation failed.

6.2.5.14 Time and time synchronisation model

6.2.5.14.1 General

The time and time synchronization test cases defined for a client device are as specified in the following text.

6.2.5.14.2 Positive test cases

The test cases listed in Table 44 shall apply.

Table 44 – Time positive test cases

Test case	Test case description
cTm1	Verify that the DUT supports and executes the SCSM time synchronisation. Change the time in the time server and verify that the DUT uses the new time.
cTm2	Check that report/logging timestamp accuracy matches the documented timestamp quality.

6.2.5.14.3 Negative test cases

The test case listed in Table 45 shall apply.

Table 45 – Time negative test cases

Test case	Test case description
cTmN1	Verify that a lost time synchronisation is detected after a specified period and the timestamp quality invalid is set.
cTmN2	Verify that the DUT handles the time stamp quality coming from the time server.

6.2.6 Acceptance criteria

Evaluation criteria for testing a Device-Under-Test (DUT) include:

- specific design characteristics to be validated,
- checkpoints identified for anomalous conditions.

There are three possibilities for a test result according to the ISO/IEC 9646 series:

- Pass (verdict) – A test verdict given when the observed test outcome gives evidence of compliance to the compliance requirement(s) on which the test purpose of the test case is focused, and when no invalid test event has been detected.
- Fail (verdict) – A test verdict given when the observed test outcome either demonstrates non-compliance with respect to (at least one of) the compliance requirement(s) on which the test purpose of the test case is focused, or contains at least one invalid test event, with respect to the relevant specification(s).
- Inconclusive (verdict) – A test verdict given when the observed test outcome is such that neither pass nor fail verdict can be given. Such a result shall be always resolved to find out if this behaviour results from the standard, from the implementation or from the test procedure.

In general, a test case is passed when the DUT behaves as specified in IEC 61400-25 (all parts) and the PIXIT, the test cases are failed when the DUT behaves different as specified in IEC 61400-25 (all parts) and PIXIT. When not specified in IEC 61400-25 (all parts) and in the PIXIT, the DUT shall keep on responding to syntactically correct messages and shall ignore syntactically incorrect messages.

7 Performance tests**7.1 General**

IEC 61400-25 (all parts) does not specify any specific performance requirements for applications operating in IEC 61400-25 (all parts) environment, but a series of essential metrics are identified. Based on this fact, Clause 7 defines the essential metrics identified within devices such that the documented product claims supporting those requirements can be compared across vendors.

7.2 Communication latency – Transfer time test introduction

The communications transfer time requirements are identified as an essential performance metric. The transfer time is the time required to deliver a process value from a sending physical device to the process logic of a receiving device. The transfer time is defined in terms of three intervals:

- t_a : the time required for the sending device to transmit the process value,
- t_b : the time required for the network to deliver the message, and
- t_c : the time required for the receiving device to deliver the value to its process logic.

The interval t_b is determined by the network infrastructure and is not an attribute of the device. From a device testing point of view, only output and input latencies can be measured, t_a and t_c are estimated from the measured latencies.

- measured output latency = estimated input processing time + estimated t_a .
- measured input latency = estimated output processing time + estimated t_b .

The vendors of network components like switches shall define and document the amount of the latency time that is due to estimated processing time for all priorities supported by the network components.

The estimated input processing time of a WPP device is the time required for input signal conditioning (e.g., debouching, sampling, etc.).

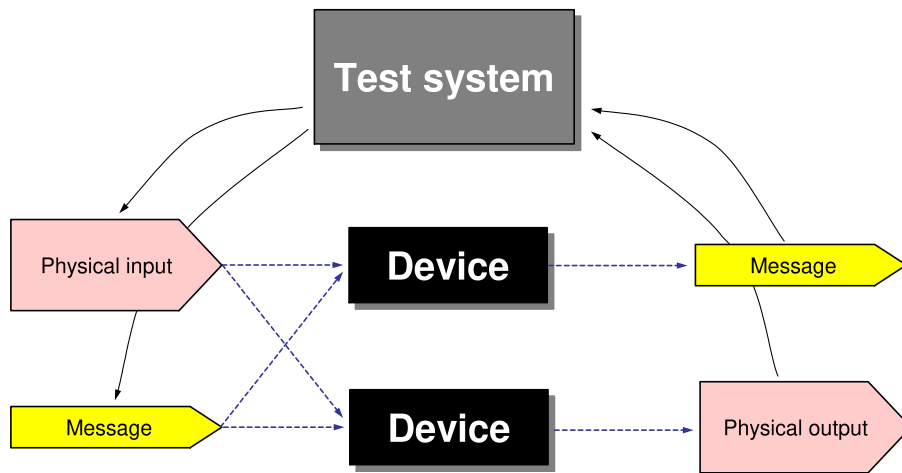
The estimated output processing time of a WPP device is the time required for output signal activation (e.g., contact delays, I/O scan rate, etc.).

The performance metrics to be measured in the WPP devices depend on which of IEC 61400-25 (all parts) services are used to deliver the process values. IEC 61400-25 (all parts) defines two mechanisms: Reporting and Control, whose transfer time can be tested.

The following time interval measurements shall be made between a physical input (or message) change and the appearance of a message on the output media (or physical output):

- report output latency;
- control output latency.

A test system (see Figure 6) shall measure an output latency time by generating a sequence of physical input triggers to the WPP device and measuring the time delay to the corresponding message generated by the device. The mean latency time and the standard deviation shall be computed across the responses to 1 000 input triggers. The vendor shall define and document the amount of the latency time which is due to estimated output processing time.



IEC

Figure 6 – Performance testing (black box principle)

The results to be documented for each latency test shall be the measured values and the two corresponding estimated values. The measured values shall be the mean values and the standard deviation of the latency time computed across 1 000 tests.

7.3 Time synchronisation and accuracy

7.3.1 Time Sync test introduction

The objective of this test is to verify the ability of the WPP device to communicate time stamp information about an instrumented event. An accurate time stamp relies on several separate functions including clock accurately decoding the received signal, accurate synchronisation of device clock to the received signal, timely device detection of change of state and accurate use of device clock value to time stamp data.

WPP devices requiring a very high time accuracy may use a directly connected external time source (radio or satellite clock).

Time synchronisation is used for the synchronisation of the device clock values when no direct external time source is available to the WPP. During synchronisation across the WPP system network, one device with a precision time source acts as the time master. A second device of the same type may be defined to act as a backup time master. The time source of the time master device is typically provided by an external source.

The time accuracy metrics defined in this subclause represent measures of time stamp accuracy for the WPP when an external source is provided or when the device relies on the time synchronisation mechanism with a time master respectively.

This test is essential due to the nature of networked WPP devices being used to design systems of interoperable devices working in a coordinated fashion. These, and other device performance measures, are essential information for predicting performance, functionality and reliability of designs executed by networked WPP devices. No specific performance benchmarks are expected to be met, however, verification and publication of actual performance measures is necessary to be conformant. Using these published performance measures, design engineers can predict the performance of the interconnected devices and thus the performance of system. Furthermore, engineers will be able to identify suitable devices for specific applications. Performance measures will be made on the device under test connected to a network with pre-defined configuration and traffic. It is understood that if the network traffic changes, the system performance may change. It is also understood that if the processing load on the device changes, the device performance may change.

7.3.2 Time Sync test methodology

7.3.2.1 General

The time synchronisation test requires a test system (see Figure 7) consisting of a data change generator function and a time master function, each connected to a common external clock source (e.g. radio or satellite clock). The change generator function triggers physical events within the WPP device, with accurate times recorded for each event. A test system analyser function retrieves the time stamp of each event from the WPP device and compares it with the recorded time of the event generation.

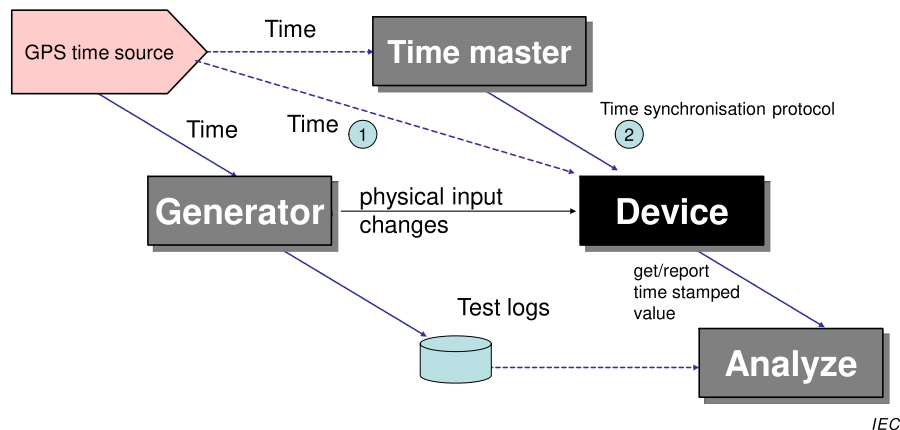


Figure 7 – Time synchronisation and accuracy test setup

7.3.2.2 Time from external source

The first accuracy measurement is made with the WPP device directly receiving time from the same external source used by the test system time master as indicated with (1) on Figure 7. A sequence of 1 000 change events is generated, and the mean and standard deviation from the mean is computed over the differences between the event times and the retrieved time stamps.

7.3.2.3 Time from time synchronisation protocol

The second accuracy measurement is made with the WPP device using the time synchronisation protocol with the Time Master function in the test system. A sequence of 1 000 change events is generated, and the mean and standard deviation from the mean is computed over the differences between the event times and the retrieved time stamps. The event sequence generation shall be coordinated with the time synchronisation protocol. The event sequence shall begin just after the device requests synchronisation with the Time Master function. If synchronisation is requested during the sequence, the sequence is interrupted while the synchronisation protocol exchange is completed.

7.3.3 Testing criteria

Time synchronisation accuracy shall be tested relative to UTC (as provided by the time reference used by the test generator).

NOTE 1 The jitter caused by network components like switches is assumed to be negligible.

The vendors of network components like switches shall define and document the amount of the latency time that is due to estimated processing time for all priorities supported by the network components. The vendors of WPP devices shall define and document the time drift of the device's internal clock.

NOTE 2 The drift is independent of the time synchronization.

Annex A (informative)

Examples of test procedure template

A.1 Example 1

sBr1	GetLogicalNodeDirectory(BRCB) and GetBRCBValues	☐ Passed ☐ Failed ☐ Inconclusive
IEC 61400-25-3, xx IEC 61400-25-4, yy		
<u>Expected result</u> DUT sends GetLogicalNodeDirectory(BRCB) Response+ DUT sends GetBRCBValues Response+		
<u>Test description</u> For each logical node Client requests GetLogicalNodeDirectory(BRCB) For each BRCB Client requests GetBRCBValues()		
<u>Comment</u> 		

A.2 Example 2

sRp1	GetLogicalNodeDirectory(URCB) and GetURCBValues	☐ Passed ☐ Failed ☐ Inconclusive
IEC 61400-25-3, xx IEC 61400-25-4, yy		
<u>Expected result</u> DUT sends GetLogicalNodeDirectory(URCB) Response+ DUT sends GetURCBValues Response+		
<u>Test description</u> For each logical node Client requests GetLogicalNodeDirectory(URCB) For each BRCB Client requests GetURCBValues()		
<u>Comment</u> 		

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